Working Notes for $D_{s1} \to D_s\gamma$ in Light-Cone QCD Sum Rules

Ds gamma project

July 2, 2026

Abstract

These notes are a living derivation for the light-cone QCD sum-rule calculation of $D_{s1}(2460) \to D_s\gamma$ and $D_{s1}(2536) \to D_s\gamma$. The purpose is pedagogical as well as practical: each step is written so that a master's student can follow the choices, conventions, and algebra before the final paper-level formulas are assembled.

1 Conventions

We use the mostly-minus metric

$$g_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \tag{1}$$

and

$$\{\gamma_\mu, \gamma_\nu\} = 2g_{\mu\nu}, \qquad \sigma_{\mu\nu} = \frac{\mathrm{i}}{2}[\gamma_\mu, \gamma_\nu], \qquad \gamma_5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3. \tag{2}$$

The Levi-Civita convention is

$$\epsilon^{0123} = +1, \qquad \epsilon_{0123} = -1, \tag{3}$$

so that

$$\mathrm{Tr}[\gamma_5\gamma_\mu\gamma_\nu\gamma_\alpha\gamma_\beta] = -4\mathrm{i}\,\epsilon_{\mu\nu\alpha\beta}. \tag{4}$$

Our Fourier convention for a propagator is

$$S(x) = \int \frac{d^4k}{(2\pi)^4}\, \mathrm{e}^{-\mathrm{i}k\cdot x}\, \widetilde{S}(k). \tag{5}$$

2 Currents and Momentum Assignment

The external momenta are

$$P = p + q, \tag{6}$$

where P is the initial axial-meson momentum, p is the final pseudoscalar D_s momentum, and q is the real-photon momentum.

The pseudoscalar interpolating current is

$$J_{D_s}(x) = \bar{s}(x)\,\mathrm{i}\gamma_5\, Q(x). \tag{7}$$

For the axial channel we use two independent currents,

$$J^{(0)}_{A\mu}(x) = \bar{s}(x)\gamma_\mu\gamma_5 Q(x), \tag{8}$$

$$J^{(0)}_{B\mu}(x) = \frac{\mathrm{i}}{m_Q + m_s}\,\bar{s}(x)\sigma_{\mu\alpha}P^\alpha\gamma_5 Q(x). \tag{9}$$

The tensor current contracts with P^α, not p^α, because it interpolates the initial axial state. The factor $1/(m_Q + m_s)$ restores the same mass dimension as the axial-vector current.

3 Correlation Function

For a generic axial current $\Gamma_\mu \in \{\gamma_\mu\gamma_5,\, \mathrm{i}\sigma_{\mu\alpha}P^\alpha\gamma_5/(m_Q+m_s)\}$ we study

$$\Pi_\mu(p,q) = \mathrm{i}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\langle\gamma(q,\varepsilon)|T\{\bar{s}(x)\Gamma_\mu Q(x)\,\bar{Q}(0)\mathrm{i}\gamma_5 s(0)\}|0\rangle. \tag{10}$$

The target Lorentz structure for the $1^+ \to 0^-\gamma$ transition is

$$(\eta\cdot\varepsilon^*)(p\cdot q) - (\eta\cdot q)(p\cdot\varepsilon^*). \tag{11}$$

On the correlator side we therefore project onto an equivalent transverse structure built from the open axial index and the photon polarization. With open indices this structure is represented by

$$T_{\mu\nu} = (p\cdot q)g_{\mu\nu} - q_\mu p_\nu, \qquad S_{\mu\nu} \equiv -T_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}. \tag{12}$$

For a real photon, $q^2=0$, it obeys

$$q^\nu S_{\mu\nu} = 0, \tag{13}$$

so the projected invariant amplitude is gauge invariant under $\varepsilon_\nu \to \varepsilon_\nu + \lambda q_\nu$. In scalar form the same statement is

$$(\eta\cdot q)(p\cdot q) - (\eta\cdot q)(p\cdot q) = 0 \tag{14}$$

after replacing $\varepsilon \to q$.

4 Wick Contraction and OPE Bookkeeping

The OPE is separated into a hard-photon sector and a soft-photon sector:

$$\Pi_\mu^{\mathrm{OPE}} = \Pi_\mu^{\mathrm{hard}} + \Pi_\mu^{\mathrm{soft}}. \tag{15}$$

This split is important because the two terms describe different physics and must not be double counted.

4.1 Hard photon emission

In the hard sector the photon is emitted perturbatively from either the heavy quark or the strange quark. After Wick contraction one obtains

$$\Pi_\mu^{\mathrm{hard}} = \mathrm{i}N_c\int d^4x\,\mathrm{e}^{\mathrm{i}p\cdot x}\mathrm{Tr}_D\Big[\Gamma_\mu S_Q^\gamma(x)\,\mathrm{i}\gamma_5\,S_s^0(-x) + \Gamma_\mu S_Q^0(x)\,\mathrm{i}\gamma_5\,S_s^\gamma(-x)\Big], \tag{16}$$

where $N_c=3$ and Tr_D denotes the Dirac trace. The free massive propagator is

$$S_q^0(x) = \int \frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\frac{\mathrm{i}(\not{k}+m_q)}{k^2-m_q^2}. \tag{17}$$

We keep explicit powers of m_s, including possible m_s^2 terms.

The first-order electromagnetic background-field correction for a generic quark flavor q is

$$S_q^\gamma(x) = -\mathrm{i}e_q\int\frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\int_0^1 du\left[\frac{\not{k}+m_q}{2(m_q^2-k^2)^2}F^{\alpha\beta}(ux)\sigma_{\alpha\beta} + \frac{ux_\alpha}{m_q^2-k^2}F^{\alpha\beta}(ux)\gamma_\beta\right]. \tag{18}$$

In Eq. (16) this generic formula is used twice:

$$S_Q^\gamma(x) = S_q^\gamma(x)\big|_{q\to Q,\; m_q\to m_Q,\; e_q\to e_Q}, \tag{19}$$

$$S_s^\gamma(x) = S_q^\gamma(x)\big|_{q\to s,\; m_q\to m_s,\; e_q\to e_s}. \tag{20}$$

Thus the hard sector contains both heavy-line and strange-line electromagnetic background-field insertions. For a real photon we write

$$F^{\alpha\beta}(ux) = \mathrm{i}(\varepsilon^\alpha q^\beta - \varepsilon^\beta q^\alpha)\mathrm{e}^{\mathrm{i}uq\cdot x}, \tag{21}$$

with the overall phase kept consistent with the Fourier convention.

Equation (18) is not a vacuum-condensate correction. It is the quark propagator in an external electromagnetic field and is included in the base analysis. What we do not include in the base hard sector are the ordinary SVZ-style condensate-expanded propagator terms such as $\langle \bar{s}s\rangle$, $\langle \bar{s}g_s\sigma Gs\rangle$, or $\langle G^2\rangle$ propagator corrections.

4.2 Equivalent momentum-space vertex form

For FeynCalc trace checks it is often simpler to use the equivalent explicit photon-vertex representation. With photon index ν,

$$N_{Q,\mu\nu} = \mathrm{Tr}_D[\Gamma_\mu(\not{k} + \not{q} + m_Q)\gamma_\nu(\not{k} + m_Q)\mathrm{i}\gamma_5(\not{k} - \not{p} + m_s)]\,, \tag{22}$$

$$N_{s,\mu\nu} = \mathrm{Tr}_D[\Gamma_\mu(\not{k} + m_Q)\mathrm{i}\gamma_5(\not{k} - \not{p} + m_s)\gamma_\nu(\not{k} - \not{p} + \not{q} + m_s)]\,. \tag{23}$$

These numerators are accompanied by the denominators

$$D_Q = [(k+q)^2 - m_Q^2][k^2 - m_Q^2][(k-p)^2 - m_s^2], \tag{24}$$

$$D_s = [k^2 - m_Q^2][(k-p)^2 - m_s^2][(k-p+q)^2 - m_s^2]. \tag{25}$$

The explicit-vertex form and the background-field form represent the same hard emission physics. We must not add both as independent contributions.

4.3 Soft photon emission: two-particle photon DAs

In the soft sector the photon is represented by its light-cone distribution amplitudes. Contracting only the heavy quark gives

$$\Pi_\mu^{\mathrm{soft},2p} = \mathrm{i}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}[\Gamma_\mu S_Q^0(x)\mathrm{i}\gamma_5]_{\alpha\beta}\langle\gamma(q,\varepsilon)|\bar{s}_\alpha(x)s_\beta(0)|0\rangle. \tag{26}$$

Fierz rearrangement converts the open spinor indices to a basis of Dirac bilinears,

$$\bar{s}_\alpha(x)s_\beta(0) = -\frac{1}{4}\sum_i(\Gamma_i)_{\beta\alpha}\bar{s}(x)\Gamma_i s(0), \tag{27}$$

so that

$$\Pi_\mu^{\mathrm{soft},2p} = -\frac{\mathrm{i}}{4}\int d^4x\, \mathrm{e}^{\mathrm{i}p\cdot x}\sum_i \mathrm{Tr}_D[\Gamma_\mu S_Q^0(x)\mathrm{i}\gamma_5\Gamma_i]\,\langle\gamma(q,\varepsilon)|\bar{s}(x)\Gamma_i s(0)|0\rangle. \tag{28}$$

The photon matrix elements are then expressed in terms of two-particle photon DAs. Parameters such as $\chi\langle\bar{s}s\rangle$ enter here as part of the photon wavefunction normalization; they are not condensate corrections to the hard propagator.

4.4 Soft photon emission: three-particle photon DAs

To include three-particle photon DAs we must also allow one background gluon to come from the heavy-quark propagator,

$$S_Q^G(x) = -\mathrm{i}g_s\int\frac{d^4k}{(2\pi)^4}\,\mathrm{e}^{-\mathrm{i}k\cdot x}\int_0^1 du\left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2}G^{\alpha\beta}(ux)\sigma_{\alpha\beta} + \frac{ux_\alpha}{m_Q^2 - k^2}G^{\alpha\beta}(ux)\gamma_\beta\right]. \tag{29}$$

This term is not a gluon-condensate correction. It is an operator insertion that combines with photon matrix elements of the form

$$\langle\gamma(q,\varepsilon)|\bar{s}(x)\Gamma_i g_s G_{\alpha\beta}(ux)s(0)|0\rangle, \tag{30}$$

which define the three-particle photon DAs.

5 Staged Calculation Plan

We proceed in two stages.

Stage 1. Include the hard photon emission terms and the two-particle photon DAs:

$$\Pi_\mu^{\text{Stage 1}} = \Pi_\mu^{\text{hard}} + \Pi_\mu^{\text{soft},2p}. \tag{31}$$

This stage keeps all explicit m_s terms and omits ordinary condensate-expanded propagators.

Stage 2. Add the three-particle photon DA contributions generated by S_Q^G and the corresponding light-cone matrix elements:

$$\Pi_\mu^{\text{Stage 2}} = \Pi_\mu^{\text{Stage 1}} + \Pi_\mu^{\text{soft},3p}. \tag{32}$$

This staging keeps the derivation auditable: first we verify the core hard and two-particle DA structures, then we add the more involved background-gluon terms.

6 Current FeynCalc Artifacts

The script `scripts/step2_current_building_blocks.wl` checks the basic Dirac conventions and defines the current vertices. The script `scripts/step2_hard_photon_numerators.wl` computes numerator-level hard-photon traces in the explicit photon-vertex representation. These traces are stored in `outputs/step2_hard_photon_numerators.txt`.

The script `scripts/step2_e1_project_hard_numerators.wl` applies the E1 projector

$$\mathcal{P}_{E1}^{\mu\nu} = \frac{p^\nu q^\mu - (p\cdot q) g^{\mu\nu}}{2(p\cdot q)^2}, \tag{33}$$

which satisfies $\mathcal{P}_{E1}^{\mu\nu}[p_\nu q_\mu - (p\cdot q)g_{\mu\nu}] = 1$ for $q^2 = 0$. The projected hard numerators are stored in `outputs/step2_e1_projected_hard_numerators.txt`.

The companion script `scripts/ward_transversality_check.wl` verifies the Ward/transversality properties of the same E1 tensor with FeynCalc. Its output is stored in `outputs/ward_transversality_check.`

The script `scripts/step2_soft_2p_fierz_traces.wl` computes the trace factors multiplying the two-particle photon DA matrix elements after the Fierz rearrangement. The output `outputs/step2_soft_2p_fierz_traces.txt` shows which bilinears feed the axial current and which feed the tensor current.

7 Stage 1 Axial-Current Numerical Baseline

Before deriving the tensor-current sum rule, we implemented the known single-axial-current result for $D_{s1}(2460) \to D_s\gamma$ as a benchmark. This is the coupling called g_1 by Colangelo, De Fazio, and Ozpineci. In our notation this is the axial-current form factor G_A.

The Stage 1 version contains

$$G_A^{\text{Stage 1}} = G_A^{\text{pert}} + G_A^{\chi\phi_\gamma} + G_A^{\mathbb{A},\bar{H}_\gamma} + G_A^{f_{3\gamma}\Psi^v}, \tag{34}$$

where the three-particle DA integral usually denoted by $\mathcal{F}_1$ is omitted until Stage 2.

The numerical script is

`scripts/stage1_axial_g1_baseline.py`.

It writes the full Borel/threshold grid to

`outputs/stage1_axial_g1_baseline.csv`.

7.1 Magnetic-susceptibility sign convention

The Colangelo paper prints $\chi = -(3.15 \pm 0.30)\,\mathrm{GeV}^{-2}$ in its matrix-element convention. In this project we store the BBK magnitude as

$$\chi(1\,\mathrm{GeV}) = +3.15 \pm 0.30\,\mathrm{GeV}^{-2}. \tag{35}$$

Therefore the sign of the $\chi\phi_\gamma$ term in the implemented sum rule is flipped relative to the literal printed formula. This is a convention translation, not a physical change. A diagnostic run that inserts positive χ into the printed negative-χ formula gives a much larger width and is kept only as a warning check.

7.2 Preliminary Stage 1 result

Using the project inputs, $u_0 = 1/2$, $3 \le M^2 \le 6\,\mathrm{GeV}^2$, and $s_0 = (2.50, 2.55, 2.60)^2\,\mathrm{GeV}^2$, the axial-current Stage 1 grid gives

$$-0.399 \le G_A^{\mathrm{Stage}\ 1} \le -0.314 \quad \mathrm{GeV}^{-1}. \tag{36}$$

With

$$\Gamma = \frac{\alpha_{\mathrm{em}}}{3} G^2 |\vec{q}_\gamma|^3, \tag{37}$$

this corresponds to

$$20.8\,\mathrm{keV} \le \Gamma_A^{\mathrm{Stage}\ 1}[D_{s1}(2460) \to D_s\gamma] \le 33.5\,\mathrm{keV}. \tag{38}$$

This is not yet the physical mixed-state result. It is the axial-current baseline that will be combined with the tensor-current form factor after the J_B sum rule is derived.

8 First Tensor-Current Step: Leading-Twist Soft Contribution

The tensor-current calculation starts from the trace-level identity produced by `scripts/step3_soft_2p_e1_pro`
For the two-particle tensor photon DA matrix element, the E1 projections are

$$\mathcal{P}_{E1}\Big[\mathrm{Tr}(J_A\, S_Q\, i\gamma_5\, \sigma_{\rho\lambda})(\varepsilon^\rho q^\lambda - \varepsilon^\lambda q^\rho)\Big] = 8\, \frac{k \cdot q}{p \cdot q}, \tag{39}$$

$$\mathcal{P}_{E1}\Big[\mathrm{Tr}(J_B\, S_Q\, i\gamma_5\, \sigma_{\rho\lambda})(\varepsilon^\rho q^\lambda - \varepsilon^\lambda q^\rho)\Big] = 8\, \frac{m_Q}{m_Q + m_s}. \tag{40}$$

After the two-particle DA Fourier integral and double Borel transform, $k = p + \bar{u}q$, hence $k \cdot q = p \cdot q$ for $q^2 = 0$. Therefore, before the hadronic decay-constant normalization, the leading-twist tensor-current OPE piece is

$$\Pi_B^{\chi\phi_\gamma} = \frac{m_Q}{m_Q + m_s}\, \Pi_A^{\chi\phi_\gamma}. \tag{41}$$

The exploratory script `scripts/stage1_tensor_gb_leading_twist.py` uses this relation to estimate only the $\chi\phi_\gamma$ part of G_B. This is not the final tensor-current sum rule. It omits the tensor-current hard contribution, the vector DA contribution, and the Stage 2 three-particle DA terms.

Using the project inputs, the partial leading-twist estimate gives

$$G_B^{\chi\phi_\gamma} \simeq +0.205 \text{ to } +0.273\ \mathrm{GeV}^{-1}, \qquad \text{if } f_B = f_T, \tag{42}$$

$$G_B^{\chi\phi_\gamma} \simeq +0.114 \text{ to } +0.151\ \mathrm{GeV}^{-1}, \qquad \text{if } f_B = f_T\, \frac{m_{D_{s1}}}{m_c + m_s}. \tag{43}$$

The sizeable sign difference relative to G_A means that the physical mixed amplitudes can be sensitive to the tensor-current normalization. This is why the next step is to finish the full G_B sum rule before quoting final mixed widths.

9 Decay-Constant Normalization Before Mixing

The labels f_A, f_B, and f_T must not be mixed casually. The axial current couples to the axial basis state through

$$\langle 0|J_A^\mu|D_{s1}^A(P,\eta)\rangle = f_A m_A \eta^\mu, \tag{44}$$

whereas our tensor-derived current is

$$J_B^\mu = \frac{\mathrm{i}}{m_c+m_s}\,\bar{s}\sigma^{\mu\alpha}P_\alpha\gamma_5 c. \tag{45}$$

The corresponding decay constant is defined by

$$\langle 0|J_B^\mu|D_{s1}^B(P,\eta)\rangle = f_B m_B \eta^\mu. \tag{46}$$

Therefore $f_B = f_T$ is not a physical statement by itself. It is only true if the tabulated f_T is already defined for the same current J_B^μ and with the same mass normalization. If instead the input table gives the more common tensor-current matrix element

$$\langle 0|\bar{s}\sigma_{\mu\nu}\gamma_5 c|D_{s1}^B(P,\eta)\rangle = f_T(\eta_\mu P_\nu - \eta_\nu P_\mu), \tag{47}$$

then contraction with $P^\nu/(m_c+m_s)$ gives the effective normalization

$$f_B^{\mathrm{eff}} \simeq f_T\,\frac{m_{D_{s1}}}{m_c+m_s}, \tag{48}$$

up to the sign fixed by the precise convention for $\sigma_{\mu\nu}\gamma_5$ and by the transverse condition $\eta\cdot P = 0$.

Only after G_A and G_B have been computed with their own basis-current decay constants do we form the physical amplitudes:

$$G_{2460} = \sin\theta\, G_A + \cos\theta\, G_B, \tag{49}$$

$$G_{2536} = \cos\theta\, G_A - \sin\theta\, G_B. \tag{50}$$

Thus the sine–cosine factors belong to the physical-state mixing of amplitudes, not to an arbitrary identification $f_B = f_T$. In the numerical scripts, $f_B = f_T$ is kept only as a diagnostic. The converted $f_B = f_T m_{D_{s1}}/(m_c+m_s)$ option is the safer central normalization unless the input-table definition of f_T is changed.

9.1 Physical-current two-point normalization check

There is one further normalization issue that should be separated from the three-point OPE calculation. The two basis currents J_A^μ and J_B^μ are not necessarily the same as the physical currents that interpolate only $D_{s1}(2460)$ or only $D_{s1}(2536)$. If we define

$$J_1^\mu = \sin\theta\, J_A^\mu + \cos\theta\, J_B^\mu, \tag{51}$$

$$J_2^\mu = \cos\theta\, J_A^\mu - \sin\theta\, J_B^\mu, \tag{52}$$

then the physical decay constants should be defined by

$$\langle 0|J_1^\mu|D_{s1}(2460)(P,\eta)\rangle = f_1 M_{2460}\eta^\mu, \tag{53}$$

$$\langle 0|J_2^\mu|D_{s1}(2536)(P,\eta)\rangle = f_2 M_{2536}\eta^\mu. \tag{54}$$

These constants are, in principle, obtained from two-point sum rules, not by identifying them directly with f_A and f_B. For the transverse invariant amplitudes of the two-point functions one has schematically

$$\Pi_{11} = \sin^2\theta\,\Pi_{AA} + \cos^2\theta\,\Pi_{BB} + 2\sin\theta\cos\theta\,\Pi_{AB}, \tag{55}$$

$$\Pi_{22} = \cos^2\theta\,\Pi_{AA} + \sin^2\theta\,\Pi_{BB} - 2\sin\theta\cos\theta\,\Pi_{AB}, \tag{56}$$

and after Borel transformation and continuum subtraction,

$$f_1^2 M_{2460}^2 \exp(-M_{2460}^2/M^2) = \widehat{\mathcal{B}}\,\Pi_{11}^{\rm OPE}, \tag{57}$$
$$f_2^2 M_{2536}^2 \exp(-M_{2536}^2/M^2) = \widehat{\mathcal{B}}\,\Pi_{22}^{\rm OPE}. \tag{58}$$

The mixed off-diagonal invariant

$$\Pi_{12} = \sin\theta\cos\theta\,(\Pi_{AA} - \Pi_{BB}) + (\cos^2\theta - \sin^2\theta)\Pi_{AB} \tag{59}$$

is a useful diagnostic: for an exactly diagonal physical-current basis it should be small in the chosen Borel window, or it can be used to determine the preferred mixing angle.

This affects only the final hadronic normalization, not the hard-photon or photon-DA OPE kernels already derived in the three-point calculation. If the three-point invariant amplitudes for the basis currents are written as

$$\Pi_A^{(3pt)} \propto f_A\,G_A, \tag{60}$$
$$\Pi_B^{(3pt)} \propto f_B\,G_B, \tag{61}$$

then the physical-current normalized couplings are

$$G_{2460}^{\rm phys} = \frac{\sin\theta\,f_A\,G_A + \cos\theta\,f_B\,G_B}{f_1}, \tag{62}$$
$$G_{2536}^{\rm phys} = \frac{\cos\theta\,f_A\,G_A - \sin\theta\,f_B\,G_B}{f_2}, \tag{63}$$

up to the obvious mass factors if the two basis-current residues are defined with different hadron-mass normalizations. The numerical tables above instead use the basis-current-normalized combination

$$G_{2460}^{\rm basis} = \sin\theta\,G_A + \cos\theta\,G_B, \tag{64}$$
$$G_{2536}^{\rm basis} = \cos\theta\,G_A - \sin\theta\,G_B. \tag{65}$$

Therefore the existing results should be kept, but labeled as the "basis-current normalization baseline". They are not invalid: they contain the completed Stage-1 and Stage-2 OPE calculation, and they are the correct starting point for phenomenology. A final physical-current normalization update requires evaluating the two-point sum rules for f_1 and f_2 using Π_{AA}, Π_{BB} and Π_{AB}. If f_1 and f_2 are close to the effective residues assumed in the basis normalization, the present tables will change only mildly; if not, the couplings in Eqs. (62)–(63) should replace the basis-current values in the final paper tables.

10 Tensor-Current Hard Term: Triangle Reduction

The hard tensor-current numerator is more complicated than the axial one because the J_B current introduces an extra factor of P^α and therefore higher powers of the loop momentum after E1 projection. The script `scripts/step3_tensor_hard_triangle_reduction.wl` rewrites scalar products such as k^2, $k\cdot p$, and $k\cdot q$ in terms of inverse propagator denominators. For the heavy-line photon emission we define

$$d_{H1} = (k+q)^2 - m_Q^2, \tag{66}$$
$$d_{H2} = k^2 - m_Q^2, \tag{67}$$
$$d_{H3} = (k-p)^2 - m_s^2, \tag{68}$$

and similarly for the strange-line photon emission. Setting $d_{Hi} = 0$ isolates the genuine three-denominator triangle core. For the tensor current this gives

$$N_{B,H}^{\text{core}} = \frac{-8\mathrm{i}m_Q^2 + 4\mathrm{i}m_Q m_s}{m_Q + m_s} - \frac{4\mathrm{i}m_Q^2 p^2}{(m_Q + m_s)(p \cdot q)}, \tag{69}$$

$$N_{B,s}^{\text{core}} = \frac{-4\mathrm{i}m_Q m_s}{m_Q + m_s} - \frac{4\mathrm{i}m_s^2 p^2}{(m_Q + m_s)(p \cdot q)}. \tag{70}$$

The same reduction was also performed for the axial current in `scripts/step3_axial_hard_triangle_redu` This is a useful calibration because the axial hard term has a known published spectral density.

10.1 Why denominator-cancellation terms are separated

Terms proportional to one or more inverse denominators cancel one or more propagators. After such a cancellation the loop integral contains at most two propagators. These reduced integrals depend on at most one of the two external virtualities, p^2 or $(p + q)^2$, and therefore do not generate the double-pole structure selected by the double Borel transform. They are recorded in the output files for auditability, but the double-pole sum rule is built from the three-denominator triangle core.

This observation narrows the remaining hard-tensor task: derive the double spectral density associated with the tensor-current triangle core and match it to the same E1 invariant amplitude used for the axial-current benchmark.

11 First Hard-Tensor Spectral Density Candidate

The next step is to translate the triangle core into a perturbative spectral density. We do this by calibrating against the known axial-current result. For one photon-emitting line, let m_i be the emitting-quark mass and m_j the spectator mass. The axial triangle core has the schematic form

$$N_i^{A,\text{core}} = -4\mathrm{i}\left[a_i + \frac{b_i}{p \cdot q}\right]. \tag{71}$$

Comparing with the published axial spectral density fixes the replacement

$$-4\mathrm{i}\, a_i \longrightarrow 2a_i \log \frac{s - m_j^2 + m_i^2 - \lambda^{1/2}(s, m_j^2, m_i^2)}{s - m_j^2 + m_i^2 + \lambda^{1/2}(s, m_j^2, m_i^2)}, \tag{72}$$

$$-4\mathrm{i}\, \frac{b_i}{p \cdot q} \longrightarrow \frac{b_i}{m_i^2} \frac{m_j^2 - m_i^2 - s}{s^2} \lambda^{1/2}(s, m_j^2, m_i^2). \tag{73}$$

The overall factor is $-3/(8\pi^2)$, and the total axial density is assembled as the strange-emission contribution minus the heavy-emission contribution, with the corresponding charges. The script

`scripts/stage1_tensor_gb_hard_candidate.py`

first applies these rules back to the axial current and reproduces Colangelo's $\rho^P(s)$ with a maximum numerical difference 1.4×10^{-17} on the test grid.

For the tensor current the core contains $p^2/(p \cdot q)$:

$$N_{B,H}^{\text{core}} = -4\mathrm{i}\left[\frac{2m_Q^2 - m_Q m_s}{m_Q + m_s} + \frac{m_Q^2 p^2}{(m_Q + m_s)(p \cdot q)}\right], \tag{74}$$

$$N_{B,s}^{\text{core}} = -4\mathrm{i}\left[\frac{m_Q m_s}{m_Q + m_s} + \frac{m_s^2 p^2}{(m_Q + m_s)(p \cdot q)}\right]. \tag{75}$$

The diagonal double-dispersion representation explains how the p^2 numerator is treated. For the double-pole part one has

$$\frac{p^2}{(s-p^2)(s-P^2)} = \frac{s}{(s-p^2)(s-P^2)} - \frac{1}{s-P^2}, \qquad P = p+q. \tag{76}$$

The second term has only a single pole and is removed by the double Borel projection in p^2 and P^2. Therefore, inside the double-pole sum rule, p^2 in the hard numerator can be replaced by the spectral variable s. The same argument gives $P^2 \to s$ for any numerator factor of P^2. The script

`scripts/step4_double_pole_p2_reduction.py`

records this identity as a short audit check.

Combining this hard candidate with the tensor-DA two-particle soft contribution gives, over the same M^2 and s_0 windows,

$$G_B^{\rm hard} \simeq -0.831 \text{ to } -0.673 \text{ GeV}^{-1}, \quad G_B^{\text{Stage 1}} \simeq -0.510 \text{ to } -0.440 \text{ GeV}^{-1}, \quad (f_B = f_T), \tag{77}$$

$$G_B^{\rm hard} \simeq -0.460 \text{ to } -0.373 \text{ GeV}^{-1}, \quad G_B^{\text{Stage 1}} \simeq -0.283 \text{ to } -0.244 \text{ GeV}^{-1}, \quad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c+m_s}\right). \tag{78}$$

At $\theta = 35.3^\circ$, the corresponding candidate mixed widths are

$$\Gamma[D_{s1}(2460)\to D_s\gamma] \simeq 62.5 \text{ to } 87.9 \text{ keV}, \quad \Gamma[D_{s1}(2536)\to D_s\gamma] \simeq 0.00 \text{ to } 0.30 \text{ keV}, \quad (f_B = f_T), \tag{79}$$

$$\Gamma[D_{s1}(2460)\to D_s\gamma] \simeq 30.9 \text{ to } 44.7 \text{ keV}, \quad \Gamma[D_{s1}(2536)\to D_s\gamma] \simeq 4.0 \text{ to } 8.2 \text{ keV}, \quad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c+m_s}\right). \tag{80}$$

These numbers should be read as a controlled Stage 1 candidate, not yet as the final result, because Stage 2 three-particle photon DAs remain. The $p^2 \to s$ step is justified for the double-pole part; any single-pole remnants belong outside the double-Borel sum rule.

12 Stage 2 Axial Three-Particle DA Check

Before deriving the tensor-current three-particle terms, we implemented the known axial-current correction as a Stage 2 calibration. The published axial sum rule contains the combination

$$\mathcal{F}_1 = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2), \tag{81}$$

integrated over the two standard regions

$$0 \le v \le 1-u_0, \qquad 0 \le \alpha_g \le \frac{u_0}{1-v}, \tag{82}$$

$$1-u_0 \le v \le 1, \qquad 0 \le \alpha_g \le \frac{1-u_0}{v}, \tag{83}$$

with

$$\alpha_q = u_0 - (1-v)\alpha_g, \qquad \alpha_{\bar q} = 1 - u_0 - v\alpha_g. \tag{84}$$

The script

`scripts/stage2_axial_g1_three_particle.py` evaluates this integral using the three-particle photon DAs in `photon_da.py`. For $u_0 = 1/2$ it finds

$$\int \mathcal{F}_1 = +7.033 \times 10^{-2}, \tag{85}$$

$$\Delta G_A^{\mathcal{F}_1} \simeq +0.0012 \text{ to } +0.0020 \text{ GeV}^{-1}. \tag{86}$$

Thus the axial Stage 2 result over the same Borel and threshold window is

$$-0.398 \leq G_A^{\text{Stage 2}} \leq -0.312 \qquad \text{GeV}^{-1}, \tag{87}$$

corresponding to

$$20.5\,\text{keV} \leq \Gamma_A^{\text{Stage 2}}[D_{s1}(2460) \to D_s\gamma] \leq 33.3\,\text{keV}. \tag{88}$$

This small correction is a useful calibration, but it does not yet complete Stage 2 for the mixed physical amplitudes. The remaining Stage 2 task is to derive the tensor-current analog of the three-particle photon DA contribution from the background-gluon insertion and project it onto the same E1 invariant.

13 First Tensor-Current Stage 2 Trace Pass

For the tensor-current three-particle term we first separated the local $\sigma_{\alpha\beta}$ part of the heavy-quark background-gluon propagator from the explicitly $x_\alpha\gamma_\beta$ part. The script

`scripts/step5_soft_3p_trace_kernels.wl`

computes the E1-projected traces for the local piece against the $\mathcal{S}$, $\tilde{\mathcal{S}}$, $\mathcal{T}_1$, and $\mathcal{T}_2$ Lorentz structures. The nonzero local kernels are

$$K_A[\mathcal{S}] = 8\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{S}] = 8\,\frac{m_Q}{m_Q+m_s}, \tag{89}$$

$$K_A[\mathcal{T}_1] = -32\mathrm{i}\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{T}_1] = -32\mathrm{i}\,\frac{m_Q}{m_Q+m_s}, \tag{90}$$

$$K_A[\mathcal{T}_2] = +32\mathrm{i}\,\frac{k\cdot q}{p\cdot q}, \qquad K_B[\mathcal{T}_2] = +32\mathrm{i}\,\frac{m_Q}{m_Q+m_s}. \tag{91}$$

The $\tilde{\mathcal{S}}$ kernel vanishes for this local part. After the three-particle light-cone Fourier integral and double Borel projection, $k\cdot q = p\cdot q$, so the local trace ratio is again

$$\frac{K_B}{K_A} = \frac{m_Q}{m_Q+m_s}. \tag{92}$$

This is the same ratio encountered in the two-particle tensor-DA family.

At this point it is useful to separate the published axial combination into the piece generated by the $\sigma_{\alpha\beta}$ part of the background-field propagator and the piece generated by the $x_\alpha G^{\alpha\beta}\gamma_\beta$ part:

$$\mathcal{F}_1^\sigma = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \tag{93}$$

$$\mathcal{F}_1^{x\gamma} = 2v\left(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2\right), \qquad \mathcal{F}_1 = \mathcal{F}_1^\sigma + \mathcal{F}_1^{x\gamma}. \tag{94}$$

The numerical split, evaluated in

`scripts/step10_stage2_f1_split.py`,

is

$$\int \mathcal{F}_1^{\sigma} = -1.5625 \times 10^{-1}, \tag{95}$$

$$\int \mathcal{F}_1^{x\gamma} = +2.2658 \times 10^{-1}, \tag{96}$$

$$\int \mathcal{F}_1 = +7.0330 \times 10^{-2}. \tag{97}$$

Thus the small published axial Stage 2 correction is partly the result of a cancellation between two larger pieces. The simple trace ratio $m_Q/(m_Q + m_s)$ is established for the $\sigma_{\alpha\beta}$ part, so the controlled sigma-matched tensor-current correction is

$$\Delta G_B^{3p} \simeq -0.00366 \text{ to } -0.00210 \text{ GeV}^{-1}, \qquad (f_B = f_T \text{ diagnostic}), \tag{98}$$

$$\Delta G_B^{3p} \simeq -0.00203 \text{ to } -0.00116 \text{ GeV}^{-1}, \qquad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s} \text{ central}\right). \tag{99}$$

The corresponding sigma-matched Stage 2 mixed-width estimate at $\theta = 35.3°$ is

$$\Gamma[D_{s1}(2460) \to D_s\gamma] \simeq 62.9 \text{ to } 88.2 \text{ keV}, \quad \Gamma[D_{s1}(2536) \to D_s\gamma] \simeq 0.00 \text{ to } 0.26 \text{ keV}, \quad (f_B = f_T \text{ diagnostic}), \tag{100}$$

$$\Gamma[D_{s1}(2460) \to D_s\gamma] \simeq 30.9 \text{ to } 44.8 \text{ keV}, \quad \Gamma[D_{s1}(2536) \to D_s\gamma] \simeq 3.8 \text{ to } 8.0 \text{ keV}, \qquad \Bigg(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s} \text{ centr} \tag{101}$$

If, instead, one multiplies the full axial $\mathcal{F}_1$ integral by the same trace ratio, one obtains only a diagnostic estimate:

$$\Delta G_B^{3p} \simeq +0.00094 \text{ to } +0.00165 \text{ GeV}^{-1}, \qquad (f_B = f_T \text{ diagnostic}), \tag{102}$$

$$\Delta G_B^{3p} \simeq +0.00052 \text{ to } +0.00091 \text{ GeV}^{-1}, \qquad \left(f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s} \text{ central}\right), \tag{103}$$

with central-normalization widths

$$\Gamma[D_{s1}(2460) \to D_s\gamma] \simeq 30.5 \text{ to } 44.5 \text{ keV}, \tag{104}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] \simeq 4.0 \text{ to } 8.1 \text{ keV}. \tag{105}$$

The difference between the sigma-matched and full-ratio diagnostic numbers is small at the current precision, but conceptually important: the $x_\alpha\gamma_\beta$ part must be matched separately before the tensor Stage 2 correction is final.

14 Stage 2 Tensor: Explicit x-Dependent Kernels

The next audit step keeps the coordinate vector x explicit. For the $\sigma_{\alpha\beta}$ part of S_Q^G, the raw E1 projections of the $\mathcal{T}_3$ and $\mathcal{T}_4$ structures are

$$K_A[\mathcal{T}_3] = +16\mathrm{i}\,\frac{k \cdot q}{p \cdot q}, \qquad K_B[\mathcal{T}_3] = +16\mathrm{i}\,\frac{m_Q}{m_Q + m_s}, \tag{106}$$

$$K_A[\mathcal{T}_4] = -16\mathrm{i}\,\frac{k \cdot q}{p \cdot q}, \qquad K_B[\mathcal{T}_4] = -16\mathrm{i}\,\frac{m_Q}{m_Q + m_s}. \tag{107}$$

Thus these two structures also obey the same ratio $K_B/K_A = m_Q/(m_Q + m_s)$ after $k \cdot q = p \cdot q$. This check is recorded in

`scripts/step6_soft_3p_x_dependent_kernels.wl`.

The genuinely new complication comes from the second term in the background propagator,

$$\frac{u\,x_\alpha}{m_Q^2-k^2}G^{\alpha\beta}(ux)\gamma_\beta. \tag{108}$$

The script

`scripts/step7_soft_3p_xgamma_kernels.wl`

computes the raw kernels with x kept explicit. Unlike the previous kernels, the tensor-to-axial ratio is not a constant at this stage. For example the $\mathcal{S}$ ratio after only $k\cdot q \to p\cdot q$ contains

$$\frac{(p\cdot q)\,(k\cdot x-p\cdot x)+(k\cdot p+p^2+2p\cdot q)(q\cdot x)}{2m_Q^2(q\cdot x)}, \tag{109}$$

and analogous expressions occur for $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$. Therefore the tensor Stage 2 three-particle result cannot yet be promoted from an estimate to a final term. The remaining calculation is to perform the Fourier/Borel replacement of the explicit $x_\alpha/(q\cdot x)$ factors, including the derivative action on the heavy-quark denominator, and then recombine the result with the photon DA structures.

As a diagnostic, we can impose only the light-cone momentum routing

$$k=p+aq, \qquad a=\alpha_{\bar q}+v\alpha_g, \qquad q^2=0, \tag{110}$$

without yet applying the derivative replacement. The script

`scripts/step8_xgamma_saddle_reduction.py`

reduces the raw ratios $K_B/[(m_Q/(m_Q+m_s))K_A]$ to

$$R_{\mathcal{S}}=\frac{p^2+(a+1)p\cdot q}{m_Q^2}, \tag{111}$$

$$R_{\mathcal{T}_1}=\frac{p^2+(3a+2)p\cdot q}{3m_Q^2}, \tag{112}$$

$$R_{\mathcal{T}_2}=\frac{p^2+(2-a)p\cdot q}{m_Q^2}, \tag{113}$$

$$R_{\mathcal{T}_3}=\frac{p^2+2a\,p\cdot q}{m_Q^2}. \tag{114}$$

If one additionally uses the heavy-line on-shell relation $m_Q^2=k^2=p^2+2a\,p\cdot q$, only $R_{\mathcal{T}_3}$ becomes exactly one. The other ratios remain nontrivial. This confirms that the $x_\alpha\gamma_\beta$ term must be treated with the full Fourier/Borel derivative machinery. It is not controlled by the simple trace-ratio argument used for the $\sigma_{\alpha\beta}$ part.

15 Stage 2 Tensor: Derivative Reduction of $x_\alpha\gamma_\beta$

The next step applies the Fourier derivative rules to the raw $x_\alpha\gamma_\beta$ kernels. With phase

$$\exp[\mathrm{i}(p+aq-k)\cdot x], \tag{115}$$

integration over x gives, up to a common phase convention,

$$p\cdot x \to -\mathrm{i}\,p\cdot\partial_k, \tag{116}$$

$$q\cdot x \to -\mathrm{i}\,q\cdot\partial_k, \tag{117}$$

$$k\cdot x\,F(k) \to -\mathrm{i}\,\partial_{k_\mu}[k_\mu F(k)] = -\mathrm{i}\,[4F(k)+k\cdot\partial_k F(k)]\,. \tag{118}$$

The script

```
scripts/step9_xgamma_derivative_reduction.wl
```

applies these rules to the scalar kernels multiplying $1/(m_Q^2 - k^2)$. After the light-cone routing $k = p + aq$, the reduced axial kernels are proportional to

$$A_{\mathcal{S}}^{x\gamma} = \frac{-8m_Q}{[-m_Q^2 + p^2 + 2a\,p\cdot q]^2}, \tag{119}$$

$$A_{\mathcal{T}_1}^{x\gamma} = \frac{48\mathrm{i}m_Q}{[-m_Q^2 + p^2 + 2a\,p\cdot q]^2}, \tag{120}$$

$$A_{\mathcal{T}_2}^{x\gamma} = \frac{16\mathrm{i}m_Q}{[-m_Q^2 + p^2 + 2a\,p\cdot q]^2}, \tag{121}$$

$$A_{\mathcal{T}_3}^{x\gamma} = \frac{-16\mathrm{i}m_Q}{[-m_Q^2 + p^2 + 2a\,p\cdot q]^2}. \tag{122}$$

The corresponding tensor-current kernels are not obtained by a universal $m_Q/(m_Q + m_s)$ multiplication. The ratios after derivative reduction are

$$R_{\mathcal{S}}^{\partial} = 1 + \frac{(1-a)p\cdot q}{m_Q^2}, \tag{123}$$

$$R_{\mathcal{T}_1}^{\partial} = \frac{2m_Q^2 + p^2 + (2-a)p\cdot q}{3m_Q^2}, \tag{124}$$

$$R_{\mathcal{T}_2}^{\partial} = -2 + \frac{3p^2 + (2+3a)p\cdot q}{m_Q^2}, \tag{125}$$

$$R_{\mathcal{T}_3}^{\partial} = \frac{2m_Q^2 - p^2 - 2a\,p\cdot q}{m_Q^2}. \tag{126}$$

All these reduced kernels carry the squared heavy-quark denominator. For the double-pole residue we drop numerator terms proportional to $D = -m_Q^2 + p^2 + 2a\,p\cdot q$, equivalently using $m_Q^2 = p^2 + 2a\,p\cdot q$ in the residue. In the three-particle integration domains,

$$a = \alpha_{\bar{q}} + v\alpha_g = 1 - u_0, \tag{127}$$

so for the symmetric Borel point $u_0 = 1/2$ this residue is evaluated at $a = 1/2$.

The script

```
scripts/step11_tensor_xgamma_ratio_integral.py
```

uses these residue ratios to match the $\mathcal{S}$, $\mathcal{T}_2$, and $\mathcal{T}_3$ pieces of the tensor-current $x_\alpha\gamma_\beta$ contribution to the known axial normalization

$$\mathcal{F}_1^{x\gamma} = 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \tag{128}$$

With the heavy-line residue condition and physical $p\cdot q$ in the E1 invariant, the matched $\mathcal{S}, \mathcal{T}_2, \mathcal{T}_3$ integral is

$$\int \mathcal{F}_{1,B}^{x\gamma}[\mathcal{S}, \mathcal{T}_2, \mathcal{T}_3] = +2.5818 \times 10^{-1}. \tag{129}$$

The tensor-current derivative reduction also produces a $\mathcal{T}_4$ residue even though the axial E1 kernel has $A_{\mathcal{T}_4}^{x\gamma} = 0$. This is not an obstacle: its normalization can be fixed directly relative to the axial $\mathcal{T}_2$ unit. On the double-pole residue,

$$\frac{B_{\mathcal{T}_4}^{x\gamma}}{[m_Q/(m_Q + m_s)]\,A_{\mathcal{T}_2}^{x\gamma}} = -\frac{(1-a)p\cdot q}{m_Q^2}. \tag{130}$$

This gives the tensor-only contribution

$$\int \mathcal{F}^{x\gamma}_{1,B}[\mathcal{T}_4] = -4.2136 \times 10^{-2}, \tag{131}$$

and therefore

$$\int \mathcal{F}^{x\gamma}_{1,B} = +2.1605 \times 10^{-1}. \tag{132}$$

Together with the sigma-like part this gives the full matched tensor three-particle integral

$$\int \left(\mathcal{F}^{\sigma}_{1,B} + \mathcal{F}^{x\gamma}_{1,B}\right) = +5.9796 \times 10^{-2}. \tag{133}$$

For comparison, the axial total is $\int \mathcal{F}_1 = +7.0330 \times 10^{-2}$. A diagnostic calculation that instead inserts the physical hadron value $p^2 = m^2_{D_s}$ directly into the OPE residue ratios gives a much larger integral, $+1.5072$, so we do not use it as the working tensor estimate.

Numerically, for the central decay-constant normalization $f_B = f_T m_{D_{s1}}/(m_c + m_s)$, the full matched Stage 2 tensor-current result gives

$$\Delta G^{3p}_B = +0.00044 \text{ to } +0.00078 \text{ GeV}^{-1}, \tag{134}$$

$$G^{\text{Stage 2}}_B = -0.2821 \text{ to } -0.2436 \text{ GeV}^{-1}. \tag{135}$$

At $\theta = 35.3^\circ$ this corresponds to

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 30.6 \text{ to } 44.5 \text{ keV}, \tag{136}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 4.0 \text{ to } 8.1 \text{ keV}. \tag{137}$$

The final matched result is close to the older full-ratio diagnostic, but the derivation is now explicit about which pieces come from the $\sigma_{\alpha\beta}$ part, which come from $x_\alpha G^{\alpha\beta}\gamma_\beta$, and how the tensor-only $\mathcal{T}_4$ term is normalized.

16 Final Stage 2 Uncertainty Scan

The script

```
scripts/final_stage2_uncertainty_scan.py
```

performs a first independent-input uncertainty scan around the completed matched Stage 2 baseline. Each random point samples the Borel parameter $M^2 \in [3, 6]$ GeV^2, the continuum threshold $\sqrt{s_0} \in [2.50, 2.60]$ GeV, and the numerical inputs listed in `inputs_table.csv`. The tensor-current decay constant is converted as

$$f_B = f_T \frac{m_{D_{s1}}}{m_c + m_s}. \tag{138}$$

The scan is not yet a correlated global error analysis; it is a transparent Monte Carlo error budget for the present sum rule. It is the preferred band to quote because it allows all inputs, M^2, and s_0 to fluctuate together.

For fixed $\theta = 35.3^\circ$, using 500 random points, the scan gives

$$G[D_{s1}(2460) \to D_s\gamma] = -0.424^{+0.065}_{-0.067} \text{ GeV}^{-1}, \tag{139}$$

$$G[D_{s1}(2536) \to D_s\gamma] = -0.140^{+0.047}_{-0.051} \text{ GeV}^{-1}, \tag{140}$$

where the uncertainty indicates the 16–84 percentile interval. The corresponding width estimates are

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 37.9^{+12.9}_{-10.7} \text{ keV}, \tag{141}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 6.1^{+5.2}_{-3.4} \text{ keV}. \tag{142}$$

If instead the mixing angle is sampled uniformly over $25^\circ \le \theta \le 45^\circ$, the same scan gives

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 36.9^{+12.6}_{-10.8}\ \text{keV}, \tag{143}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 6.0^{+7.6}_{-4.1}\ \text{keV}. \tag{144}$$

The full scan table is written to

`outputs/final_stage2_uncertainty_scan.csv`.

For visualization the script

`scripts/uncertainty_publication_plots.py`

also fits Gaussian curves to the Monte Carlo width histograms using the sample mean and sample standard deviation. These fits are descriptive rather than the primary quoted uncertainties, because the width distribution is mildly right-tailed, especially for $D_{s1}(2536)$. For the fixed-angle ensemble the Gaussian summaries are

$$\Gamma_{2460} = 38.99 \pm 12.76\ \text{keV}, \tag{145}$$

$$\Gamma_{2536} = 7.39 \pm 5.67\ \text{keV}, \tag{146}$$

while for $25^\circ \le \theta \le 45^\circ$ they are

$$\Gamma_{2460} = 38.47 \pm 12.78\ \text{keV}, \tag{147}$$

$$\Gamma_{2536} = 7.79 \pm 6.76\ \text{keV}. \tag{148}$$

The corresponding publication histograms are shown in Figs. 1 and 2.

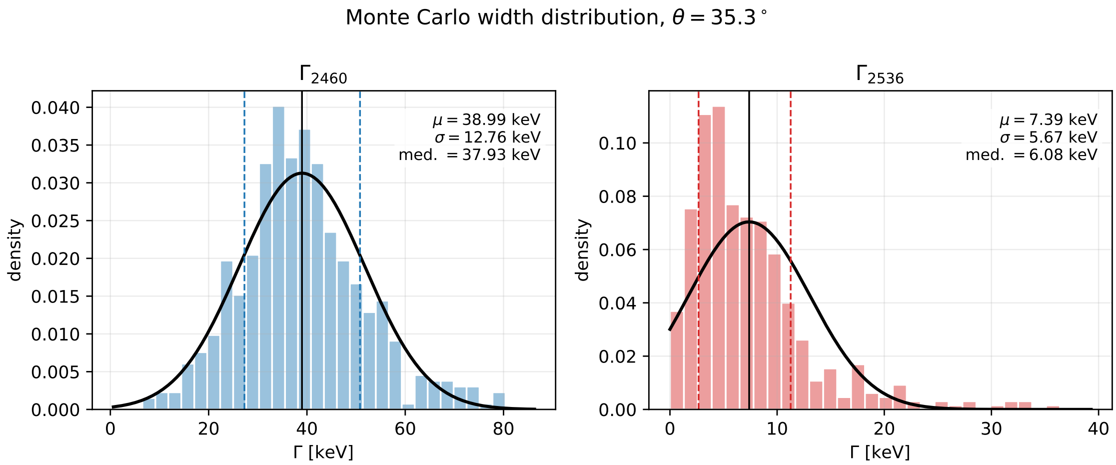

Figure 1: Monte Carlo width distributions for fixed $\theta = 35.3^\circ$. The black curves are Gaussian overlays using the sample mean and standard deviation; the dashed vertical lines mark the 16–84 percentile interval.

The diagnostic script

`scripts/uncertainty_budget_one_at_time.py`

then varies one input group at a time around the central point. This is not the band to quote as the final uncertainty; it explains which input groups drive the Monte Carlo spread.

The main lesson from Table 1 is that the $D_{s1}(2460)$ width is most sensitive to condensate/χ inputs and decay constants, while the $D_{s1}(2536)$ width is particularly sensitive to the mixing angle because the physical amplitude contains a cancellation between the axial and tensor-current components. The same ranking is visualized in Fig. 3.

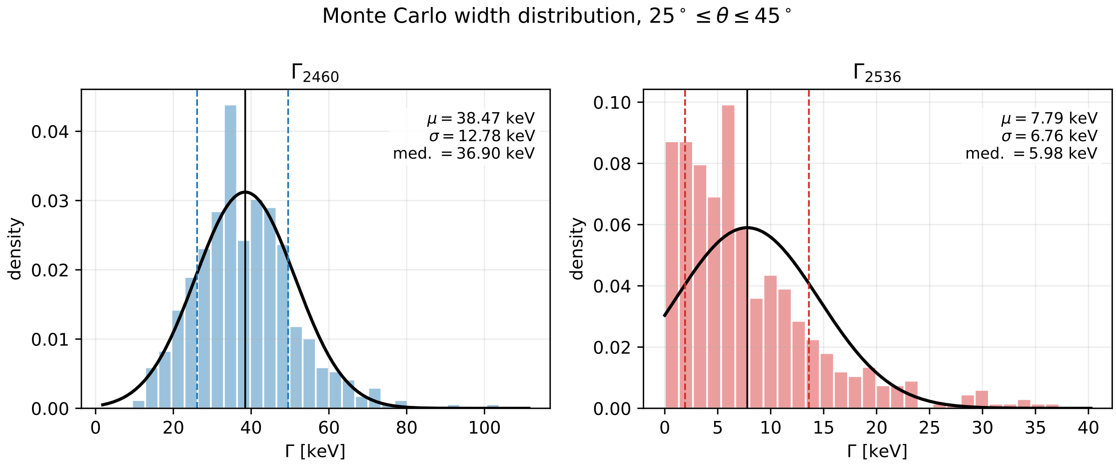

Figure 2: Monte Carlo width distributions when the mixing angle is sampled uniformly over $25^\circ \leq \theta \leq 45^\circ$. The black curves are Gaussian overlays using the sample mean and standard deviation; the dashed vertical lines mark the 16–84 percentile interval.

Varied group	Γ_{2460} (keV)	Γ_{2536} (keV)
$M^2, \sqrt{s_0}$	37.4 [33.0, 41.1]	5.95 [5.05, 7.00]
θ	37.3 [33.7, 39.6]	5.83 [2.40, 11.2]
quark and hadron masses	37.1 [35.1, 39.1]	6.01 [4.92, 7.27]
decay constants	37.7 [32.9, 42.4]	5.91 [3.48, 9.42]
condensates and χ	38.7 [29.2, 48.6]	6.40 [4.00, 9.11]
photon DA shape	37.2 [37.2, 37.2]	6.04 [6.04, 6.04]
all inputs, fixed θ	37.4 [26.8, 50.2]	5.50 [2.73, 11.2]
all inputs and θ	36.1 [25.7, 48.3]	6.28 [1.78, 14.9]

Table 1: One-at-a-time 16–84 percentile uncertainty budget with 500 random points per group. The photon-shape row is flat in the present matched reduction; it should be read as negligible sensitivity in this implementation, not as a general statement about photon DAs.

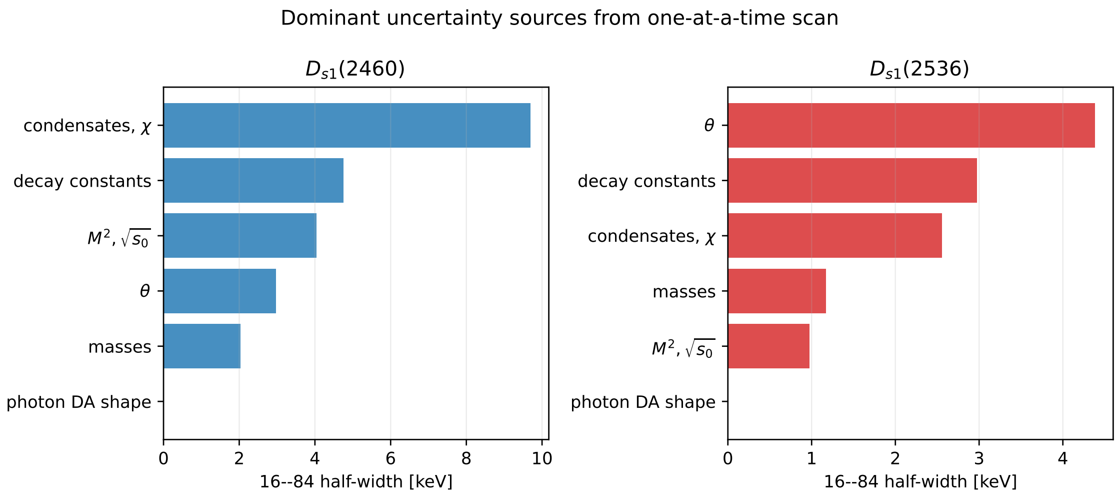

Figure 3: Ranked one-at-a-time uncertainty sources. The plotted quantity is half of the 16–84 percentile width for each group.

16.1 Alternative lattice normalization of the leading-twist photon DA

The dominant uncertainty in the legacy scan is the combination $\chi\langle\bar{s}s\rangle$ entering the leading-twist photon DA. The Rohrwild/BBK-style baseline treats χ and $\langle\bar{s}s\rangle$ as separate inputs. A recent lattice calculation by Bacchio, Becirevic, Gagliardi, and Sanfilippo gives the strange-quark normalization directly,

$$f^{\perp}_{\gamma,s}(2\text{ GeV}) = -51.0(1.8)\text{ MeV}, \tag{149}$$

where

$$f^{\perp}_{\gamma,s}(\mu) = \chi_s(\mu)\langle\bar{s}s\rangle(\mu). \tag{150}$$

To compare with the working $\mu = 1$ GeV setup, the script

`scripts/lattice_photon_normalization_comparison.py`

uses a leading-order tensor-current running factor $f^{\perp}_{\gamma,s}(1\text{ GeV})/f^{\perp}_{\gamma,s}(2\text{ GeV}) = 1.08$, giving

$$f^{\perp}_{\gamma,s}(1\text{ GeV}) = -55.1(1.9)\text{ MeV}. \tag{151}$$

Only the leading-twist normalization is replaced in this comparison; the higher-twist and three-particle terms that contain $\langle\bar{s}s\rangle$ remain explicit. This makes the comparison conservative and easy to audit.

Input scenario	Γ_{2460} (keV)	Γ_{2536} (keV)
Legacy $\chi\langle\bar{s}s\rangle$, fixed θ	$36.7\,[26.8, 50.2]$	$5.60\,[2.56, 11.0]$
Lattice $f^{\perp}_{\gamma,s}$, fixed θ	$14.4\,[11.4, 18.8]$	$0.68\,[0.12, 1.88]$
Legacy $\chi\langle\bar{s}s\rangle$, θ scan	$36.3\,[24.8, 50.2]$	$5.74\,[1.86, 12.9]$
Lattice $f^{\perp}_{\gamma,s}$, θ scan	$14.6\,[11.3, 18.6]$	$0.64\,[0.06, 2.50]$

Table 2: Comparison of the legacy photon-DA normalization with the lattice-normalized leading-twist photon DA. The intervals are 16–84 percentiles from 500 random points per ensemble.

The lattice-normalized scenario substantially reduces the uncertainty and also shifts the central widths downward. This should be presented as an important alternative input scenario rather than folded silently into the legacy result, because it changes the central leading-twist normalization. The corresponding histograms are shown in Fig. 4.

For direct comparison with the legacy Monte Carlo histograms in Figs. 1 and 2, the same script also produces Gaussian-overlay plots for the lattice-normalized scenario. For fixed $\theta = 35.3°$, the descriptive Gaussian summaries are

$$\Gamma_{2460}^{\text{lat. } f^{\perp}_{\gamma}} = 14.91 \pm 3.69\text{ keV}, \tag{152}$$

$$\Gamma_{2536}^{\text{lat. } f^{\perp}_{\gamma}} = 0.97 \pm 0.97\text{ keV}. \tag{153}$$

For the mixing-angle scan they are

$$\Gamma_{2460}^{\text{lat. } f^{\perp}_{\gamma}} = 14.94 \pm 3.77\text{ keV}, \tag{154}$$

$$\Gamma_{2536}^{\text{lat. } f^{\perp}_{\gamma}} = 1.21 \pm 1.45\text{ keV}. \tag{155}$$

The $D_{s1}(2536)$ distribution is visibly non-Gaussian near zero because the amplitude is cancellation-driven, so the percentile intervals in Table 2 remain the safer quoted uncertainties.

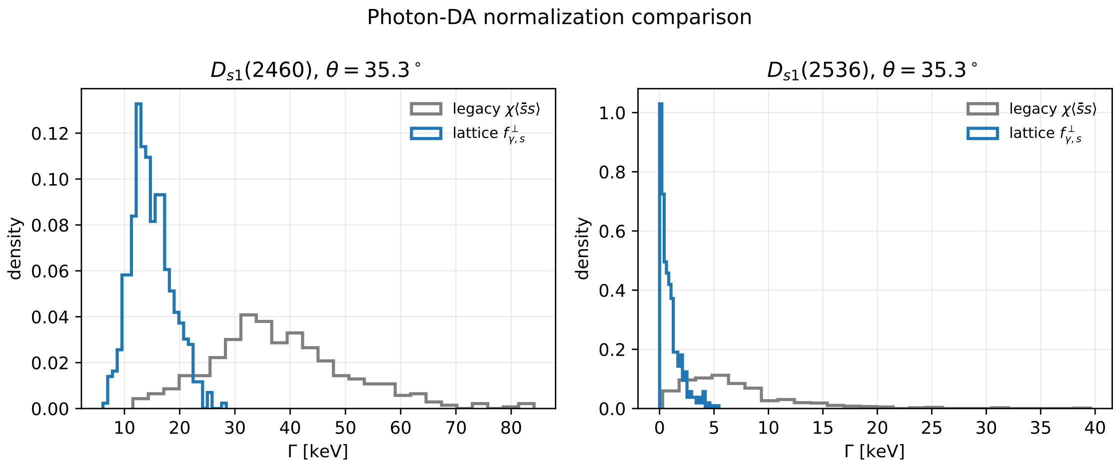

Figure 4: Fixed-angle Monte Carlo comparison between the legacy $\chi\langle\bar{s}s\rangle$ treatment and the lattice-normalized $f^{\perp}_{\gamma,s}$ treatment.

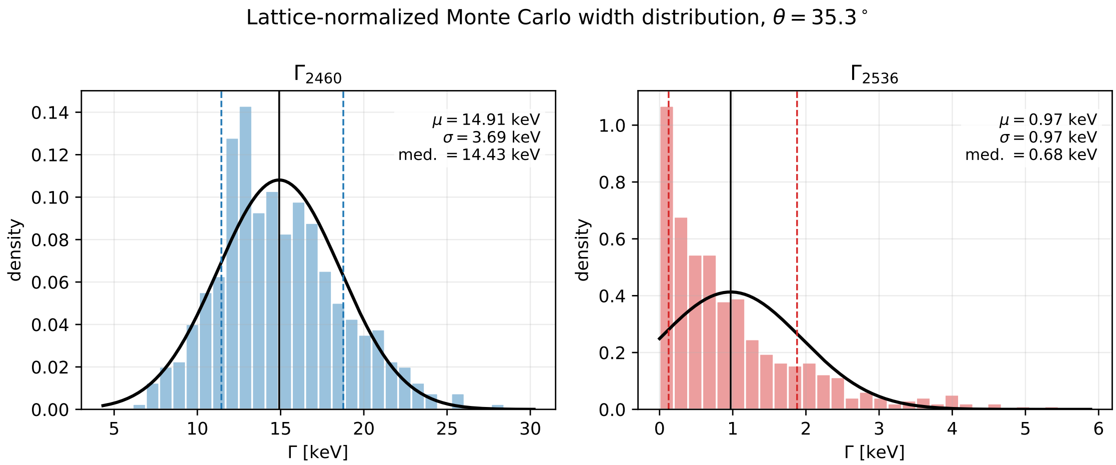

Figure 5: Lattice-normalized Monte Carlo width distributions for fixed $\theta = 35.3°$. The black curves are Gaussian overlays using the sample mean and standard deviation; dashed vertical lines mark the 16–84 percentile interval.

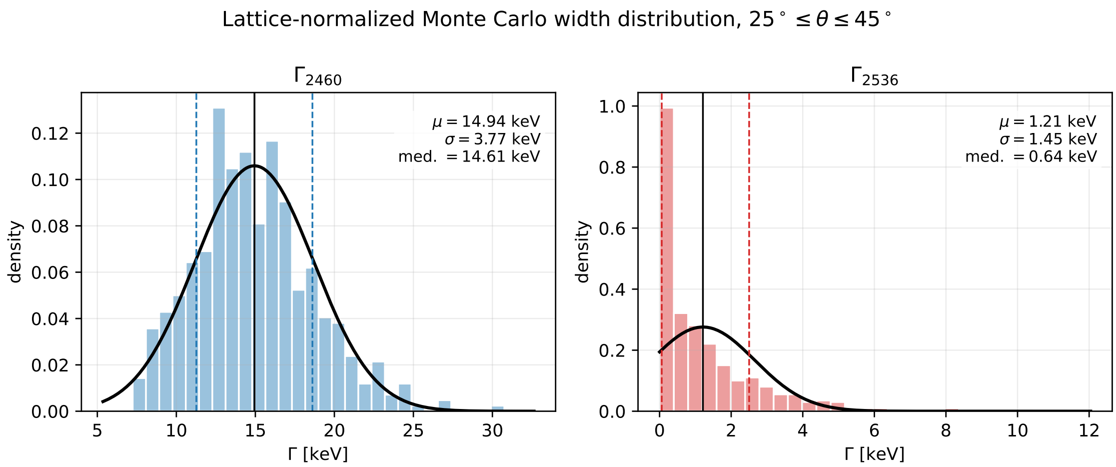

Figure 6: Lattice-normalized Monte Carlo width distributions when the mixing angle is sampled over $25° \leq \theta \leq 45°$.

16.2 Recommended D_{s1} results table

For the manuscript, the safest way to report the width predictions is not a Gaussian mean and standard deviation, but the median with asymmetric 16–84 percentile intervals. This is especially important for $D_{s1}(2536)$, where the width is positive definite and the amplitude can pass close to zero because of the axial–tensor cancellation. Table 4 now uses the Stage-3 physical-current residue normalization. In this first implementation the off-diagonal two-point overlap is calibrated from the working Stage-3 anchor $f_1 = 0.345 \pm 0.017$ GeV, and the same overlap then predicts f_2. Numerically this gives $f_2 \simeq 0.38$ GeV and a negative overlap $\rho_{AB} \simeq -0.4$. The previous basis-current table remains useful as a pre-residue diagnostic, but it should no longer be used as the headline D_{s1} result.

As a separate verification check, we compared the mixed-current residue model with the pure axial-current local-QCDSR benchmark in Wang's Table 1, $f_A(D_{s1}) = 0.245 \pm 0.017$ GeV. This benchmark is not treated as an authority for the physical mixed residues f_1 and f_2, because it uses the pure current $\bar{Q}\gamma_\mu\gamma_5 q$. Therefore the comparison must be made only in a limiting-current configuration where one of our physical currents becomes J_A. The model gives $f_1 = f_A$ at $\theta = 90°$, where $J_1 = J_A$, and $f_2 = f_A$ at $\theta = 0°$, where $J_2 = J_A$. Our working pure-axial input, $f_A = 0.225 \pm 0.025$ GeV, is close to Wang's 0.245 ± 0.017 GeV and agrees within the combined uncertainty; the small central-value difference is naturally attributable to different quark masses, condensates, continuum thresholds, Borel windows, OPE truncations, and normalization conventions. At the physical angle $\theta = 35.3°$, the Wang-like pure-axial target lies near the lower edge of the mixed-current residue range and requires a strongly negative overlap, $\rho_{AB} \simeq -1$ with our working $f_A = 0.225$ GeV, or $\rho_{AB} \simeq -0.96$ if the Wang value is also used as the basis f_A. By contrast, the headline Stage-3 anchor $f_1 = 0.345$ GeV corresponds to $\rho_{AB} \simeq -0.41$ and $f_2 \simeq 0.381$ GeV. The check is stored in `outputs/stage3_residue_benchmark_checks.csv`.

Case	Current/target	θ	ρ_{AB}	f_A	f_1	f_2
Wang benchmark	pure J_A two-point current	–	–	0.245 ± 0.017	–	–
our pure-J_1 axial limit	$J_1 = J_A$	90°	0	0.225	0.225	0.462
our pure-J_2 axial limit	$J_2 = J_A$	0°	0	0.225	0.462	0.225
physical angle, Wang-like target	target $f_1 = 0.245$	35.3°	-1.00	0.225	0.247	0.451
physical angle, Wang f_A	target $f_1 = f_A = 0.245$	35.3°	-0.957	0.245	0.245	0.462
current Stage-3 anchor	target $f_1 = 0.345$	35.3°	-0.408	0.225	0.345	0.381

Table 3: Comparison of Wang's pure axial-current benchmark with limiting cases of the present mixed-current residue model. Wang's number checks the pure J_A scale, so the appropriate comparison is the pure-J_1 axial limit at $\theta = 90°$ or the pure-J_2 axial limit at $\theta = 0°$. The physical mixed-current residues require assumptions about the AA/BB overlap and are therefore not directly determined by Wang's pure axial-current benchmark. All decay constants are in GeV. A machine-readable version is `outputs/wang_pure_axial_residue_comparison_table.csv`.

There is also a scale-convention issue that should be stated explicitly. The photon-DA normalizations in the table are quoted at $\mu = 1$ GeV, whereas the quark masses used in the hard amplitudes are the usual $\overline{\text{MS}}$ inputs, e.g. $m_s(2\text{ GeV})$ and $m_c(m_c)$. This is a standard phenomenological input convention, but it is not a fully matched single-scale setup. A strict scale-consistent variant would evolve m_s, $\langle \bar{s}s \rangle$, and either χ_s or $f^{\perp}_{\gamma,s}$ to a common factorization scale before rerunning the sum rule. We therefore flag the present rows as mixed-scale inputs rather than interpreting them as a separate scale-consistent prediction.

The legacy rows should be retained for comparison with the older photon-DA literature. The lattice-normalized rows are the more modern input scenario and give a much tighter prediction for $D_{s1}(2460)$, while $D_{s1}(2536)$ remains best described by an asymmetric interval because the central amplitude is cancellation-suppressed. The machine-readable version of Table 4 is

Scenario	Photon input	implied χ_s	Mixing input	Scale check	Γ_{2460} (keV)	Γ_{2536} (keV)
Legacy	$\chi_s(1\text{ GeV}) = 3.15 \pm 0.30$ (Rohrwild/BBK)	$3.15\,[2.89, 3.47]$	$\theta = 35.3^\circ$	mixed-scale input	$37.7^{+11.5}_{-10.5}$	$0.045^{+0.124}_{-0.040}$
Legacy	$\chi_s(1\text{ GeV}) = 3.15 \pm 0.30$ (Rohrwild/BBK)	$3.16\,[2.86, 3.42]$	$25^\circ \leq \theta \leq 45^\circ$	mixed-scale input	$37.8^{+10.3}_{-11.5}$	$0.350^{+0.752}_{-0.312}$
Lattice	$f^{\perp}_{\gamma,s}(1\text{ GeV}) = -55.1 \pm 1.9$ MeV (Bacchio et al.)	$4.99\,[4.24, 6.05]$	$\theta = 35.3^\circ$	mixed-scale input	$16.1^{+4.4}_{-3.7}$	$0.504^{+0.600}_{-0.377}$
Lattice	$f^{\perp}_{\gamma,s}(1\text{ GeV}) = -55.1 \pm 1.9$ MeV (Bacchio et al.)	$5.02\,[4.25, 6.09]$	$25^\circ \leq \theta \leq 45^\circ$	mixed-scale input	$15.8^{+5.0}_{-3.7}$	$0.498^{+1.087}_{-0.457}$

Table 4: Recommended reporting table for the $D_{s1} \to D_s\gamma$ widths. Entries are medians with 16–84 percentile intervals. The lattice scenario replaces only the leading-twist normalization $\chi_s\langle\bar{s}s\rangle$ by $f^{\perp}_{\gamma,s}$; higher-twist condensate terms are left explicit. The implied χ_s column is derived from $f^{\perp}_{\gamma,s}/\langle\bar{s}s\rangle$ in the lattice rows, so it inherits the strange-condensate spread. The scale-check column records that the present numerical rows use conventional mixed-scale phenomenological inputs. The widths include the Stage-3 physical-current residue correction calibrated with the working Stage-3 anchor $f_1 = 0.345 \pm 0.017$ GeV; a fully matched-scale prediction should be generated as a separate rerun.

`outputs/ds1_recommended_results_table.csv`. The full Stage-3 residue-normalization scan is stored in `outputs/stage3_ds1_physical_residue_scan.csv`, with the compact summary in `outputs/stage3_ds1_physical_residue_summary.csv`.

17 Central Borel Stability

For the central input set and $\theta = 35.3^\circ$, the script

`scripts/stage2_stability_plots.py`

produces the Borel-stability curves shown in Fig. 7. The curves are smooth over $3 \leq M^2 \leq 6\text{ GeV}^2$. At fixed $\sqrt{s_0}$, the dependence on M^2 is mild; changing $\sqrt{s_0}$ from 2.50 to 2.60 GeV shifts the normalization more visibly and should therefore remain part of the quoted systematic uncertainty.

Across the full displayed window the central ranges are

$$\Gamma[D_{s1}(2460) \to D_s\gamma] = 30.6 \text{ to } 44.5 \text{ keV}, \tag{156}$$

$$\Gamma[D_{s1}(2536) \to D_s\gamma] = 3.95 \text{ to } 8.10 \text{ keV}. \tag{157}$$

The corresponding amplitude spans are smaller:

$$G_{2460} = -0.460 \text{ to } -0.381\text{ GeV}^{-1}, \tag{158}$$

$$G_{2536} = -0.162 \text{ to } -0.113\text{ GeV}^{-1}. \tag{159}$$

This is the expected pattern: the widths amplify the residual sum-rule variation because $\Gamma \propto G^2|\vec{q}|^3$.

18 Publication Figures and Phenomenology

The remaining publication-oriented figures are generated by

`scripts/publication_plots.py`.

Figure 8 shows the continuum-threshold dependence at fixed representative values of M^2. It complements Fig. 7: the curves are monotonic in $\sqrt{s_0}$, so the continuum threshold should be quoted as an important systematic.

Figure 9 shows the mixing-angle dependence. The $D_{s1}(2460)$ width remains at the tens-of-keV level over a broad range of θ, while the $D_{s1}(2536)$ width is strongly suppressed near the

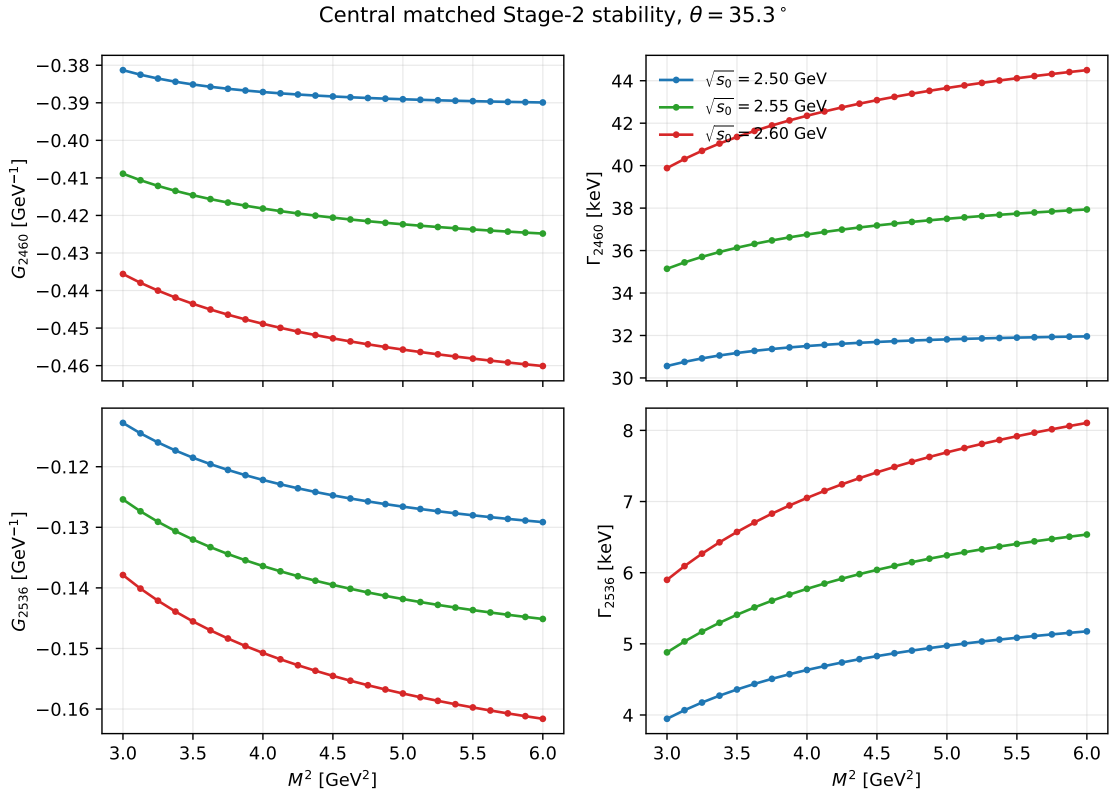

Figure 7: Central matched Stage 2 stability for $D_{s1}(2460) \to D_s\gamma$ and $D_{s1}(2536) \to D_s\gamma$ at $\theta = 35.3°$. The three curves correspond to $\sqrt{s_0} = 2.50, 2.55, 2.60$ GeV.

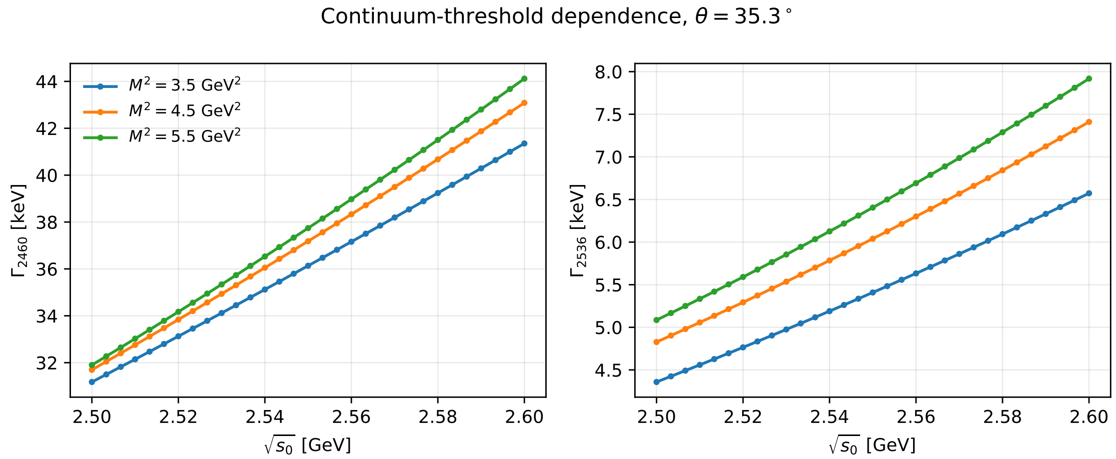

Figure 8: Continuum-threshold dependence of the matched Stage 2 widths for $\theta = 35.3°$ at three representative Borel points.

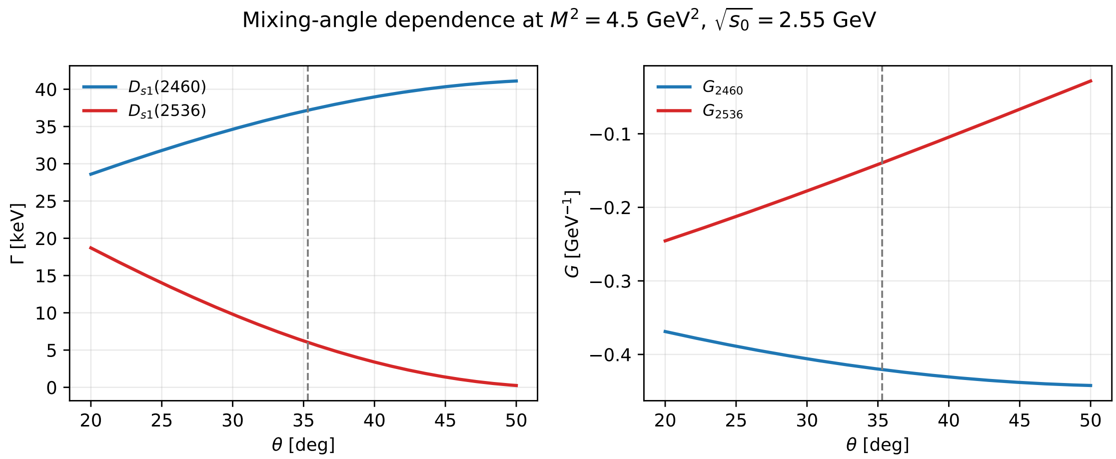

Figure 9: Mixing-angle dependence at $M^2 = 4.5$ GeV^2 and $\sqrt{s_0} = 2.55$ GeV. The dashed line marks $\theta = 35.3°$.

large-angle region because its amplitude involves a cancellation between the axial and tensor-current components.

The cancellation pattern is displayed more explicitly in Fig. 10. At the central point $M^2 = 4.5$ GeV^2, $\sqrt{s_0} = 2.55$ GeV, the $D_{s1}(2536)$ amplitude receives opposite-sign contributions from the axial Stage 1 term and the hard tensor-current term. The three-particle DA corrections are numerically small in the final physical amplitudes, but they are included in the quoted Stage 2 result.

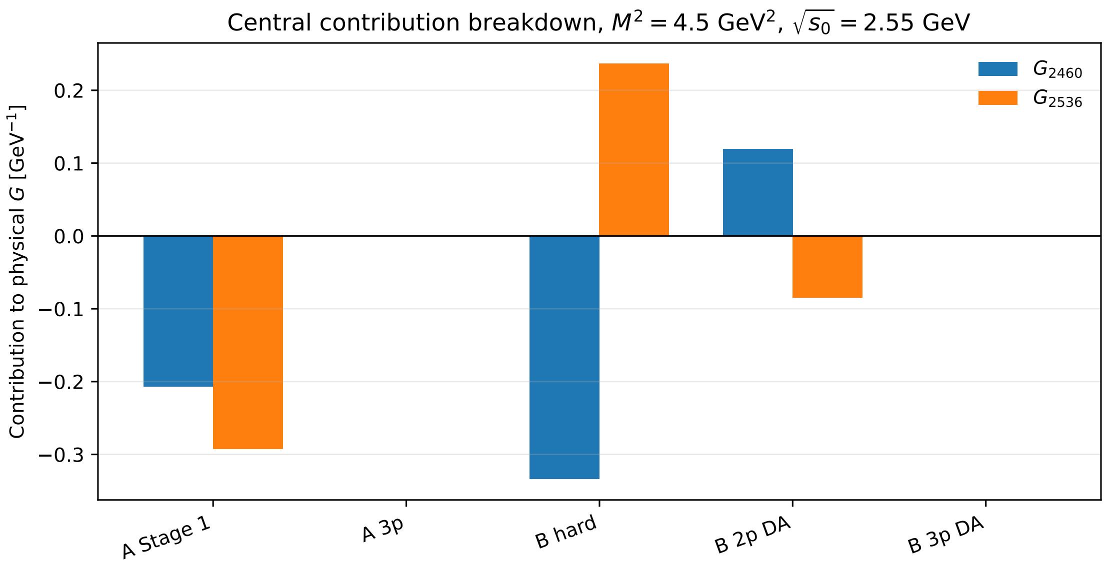

Figure 10: Contribution breakdown for the physical couplings at the central Borel and threshold point. The labels "A" and "B" refer to the axial-current and tensor-current components before physical mixing.

Finally, Fig. 11 compares the present result for $D_{s1}(2460) \to D_s\gamma$ with representative estimates collected by Colangelo, De Fazio and Ozpineci. The present result is higher than the original axial-current LCSR interval, but it remains the same order of magnitude. It should not be advertised as explaining a non-observation by making the $D_{s1}(2460) \to D_s\gamma$ width tiny. Instead, the robust phenomenological statement from this calculation is that the $D_{s1}(2536) \to D_s\gamma$

channel is much more cancellation-sensitive and can be substantially smaller.

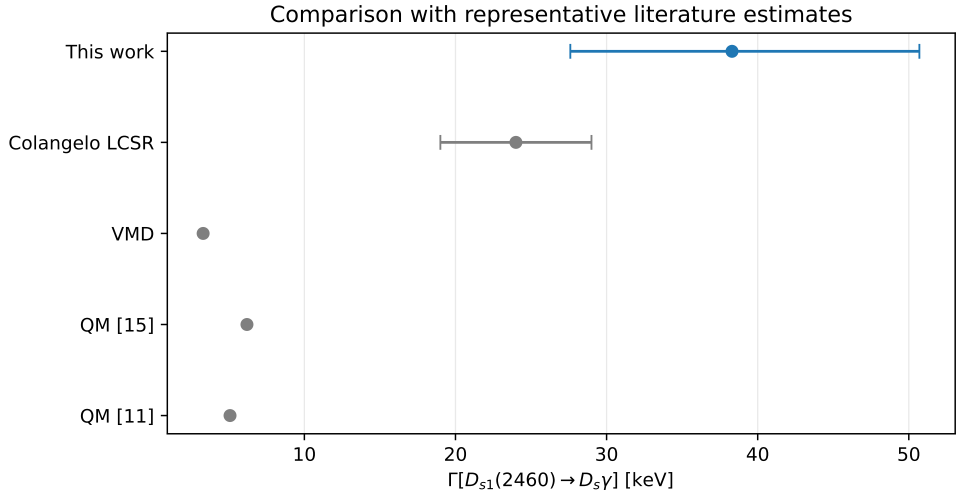

Figure 11: Comparison of the present $D_{s1}(2460) \to D_s\gamma$ width with representative literature estimates. The Colangelo LCSR, VMD and quark-model values are from the comparison table in Colangelo, De Fazio and Ozpineci (hep-ph/0505195).

19 Validation Checks Before the B_{s1} Extension

The script

`scripts/validation_checks.py`

collects several checks that should be repeated whenever the numerical implementation is modified. The first check is an axial-current benchmark against Colangelo, De Fazio and Ozpineci, using paper-like input choices and the BBK value $\kappa = 0.2$ for the relevant photon DA parameter. Over $3 \le M^2 \le 6$ GeV^2 and $2.50 \le \sqrt{s_0} \le 2.60$ GeV, the present controlled benchmark gives

$$g_1 = -0.340 \text{ to } -0.237 \text{ GeV}^{-1}, \qquad \Gamma = 11.8 \text{ to } 24.3 \text{ keV}. \tag{160}$$

The published comparison band is $g_1 = -0.37$ to -0.29 GeV^{-1} and $\Gamma = 19$ to 29 keV. This is close enough in sign and scale to validate the implementation, but it also shows that an exact literature reproduction requires matching every convention and input choice. We should present this as a benchmark check, not as an independent final phenomenological result.

The same validation script checks the photon DA normalization and the three-particle integration closure:

$$\int_0^1 du\, \varphi_\gamma(u) = 1.0000000000, \tag{161}$$

$$\max_u |\psi^a(u) - \psi^a(1-u)| = 4.44 \times 10^{-15}, \tag{162}$$

$$\max_u |\psi^v(u) + \psi^v(1-u)| = 1.78 \times 10^{-15}, \tag{163}$$

$$F_1 - F_1^{(1)} - F_1^{(2)} = 0. \tag{164}$$

These are numerical quadrature checks of the DA implementation and the split integration domains.

The independent FeynCalc script `scripts/ward_transversality_check.wl` checks the selected E1 tensor $S_{\mu\nu} = p_\nu q_\mu - (p\cdot q)g_{\mu\nu}$. With $q^2 = 0$, $\varepsilon \cdot q = 0$, $P = p + q$, and $\eta \cdot P = 0$, it gives

$$S_{\mu\nu} q^\nu = 0, \tag{165}$$

$$[(\eta\cdot\varepsilon)(p\cdot q) - (\eta\cdot q)(p\cdot\varepsilon)]_{\varepsilon\to q} = 0, \tag{166}$$

$$P^\mu S_{\mu\nu}\varepsilon^\nu = 0, \tag{167}$$

$$\frac{S_{\mu\nu}S^{\mu\nu}}{2(p\cdot q)^2} = 1. \tag{168}$$

The first two equations are the photon Ward check, the third confirms compatibility with the physical initial axial-meson polarization, and the last line is the projector normalization.

At the central point $M^2 = 4.5$ GeV^2 and $\sqrt{s_0} = 2.55$ GeV, the hard tensor-current charge decomposition is

$$G_B^{\text{hard}} = -4.092 \times 10^{-1}\ \text{GeV}^{-1}, \tag{169}$$

$$G_B^{\text{hard}}(e_c\ \text{line}) = -2.499 \times 10^{-1}\ \text{GeV}^{-1}, \tag{170}$$

$$G_B^{\text{hard}}(e_s\ \text{line}) = -1.594 \times 10^{-1}\ \text{GeV}^{-1}. \tag{171}$$

This confirms that hard photon emission from both quark lines is numerically important and must remain in the baseline.

Diagnostic case	G_A (GeV^{-1})	G_B	Γ_{2460} (keV)	Γ_{2536}
central	-0.357	-0.263	37.18	6.04
$m_s \to 0$ numerical proxy	-0.175	-0.210	15.59	0.15
$m_s = m_c$ toy	$+0.535$	$+0.273$	59.46	24.22
$e_c = 0$ toy	$+0.102$	-0.013	0.50	2.57
$e_s = 0$ toy	-0.459	-0.250	46.30	16.48
light-u OPE toy	-1.040	-0.438	192.96	110.02
light-d OPE toy	-0.109	-0.176	8.94	0.05

Table 5: Limiting-case diagnostics from `outputs/validation_limiting_cases.csv`. The $m_s \to 0$ and light-quark rows use $m_q = 10^{-4}$ GeV as a numerical regulator of the present hard-line spectral-density representation. The light-u and light-d rows are OPE-level charge and condensate substitutions only, not physical nonstrange $D_1 \to D\gamma$ predictions.

20 First $B_{s1} \to B_s\gamma$ Extension

The same matched Stage-2 machinery can be reused for the bottom-strange system after the replacement $m_c \to m_b$ and $e_c \to e_b = -1/3$. The script

`scripts/bs1_stage2_baseline.py`

performs this first pass. Internally the reusable functions still use the key `mc`; in this section it should be read as m_b.

The first-pass numerical inputs are deliberately conservative working inputs: $m_b(m_b) = 4.18$ GeV, $m_s(2\ \text{GeV}) = 0.093$ GeV, $m_{B_s} = 5.36692$ GeV, and $f_{B_s} = 0.2303$ GeV (FLAG Review 2024). For the axial and tensor B_{s1} decay constants we use the temporary heavy-quark scaling estimate $f_H\sqrt{m_H} = \text{const.}$ from the D_{s1} central values. Thus these rows are not yet final paper predictions; they are a controlled baseline for the bottom-sector calculation.

Two mass targets are shown. The observed narrow state is treated as the high-combination state,

$$G_{\text{high}} = \cos\theta\, G_A - \sin\theta\, G_B, \qquad m_{B_{s1}} = 5.82870 \text{ GeV}. \tag{172}$$

Because the lower 1^+ bottom-strange partner is not firmly established, the second row uses $m_{B_{s1}} = 5.720$ GeV only as a lower-partner diagnostic, with

$$G_{\text{low}} = \sin\theta\, G_A + \cos\theta\, G_B. \tag{173}$$

The scan uses $8 \le M^2 \le 14$ GeV^2, $\theta = 35.3^\circ$, and $\sqrt{s_0} = 6.05, 6.15, 6.25$ GeV for the observed state ($5.95, 6.05, 6.15$ GeV for the lower diagnostic state).

Target	Photon input	G (GeV^{-1})	Γ (keV)
$B_{s1}(5830)$ high combo	legacy χ_s (Rohrwild/BBK)	$0.0358\,[0.0218, 0.0688]$	$0.27\,[0.10, 1.01]$
$B_{s1}(5830)$ high combo	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$0.1496\,[0.119, 0.216]$	$4.75\,[3.02, 9.88]$
lower 1^+ diagnostic	legacy χ_s (Rohrwild/BBK)	$0.589\,[0.502, 0.761]$	$33.8\,[24.6, 56.5]$
lower 1^+ diagnostic	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$0.847\,[0.725, 1.09]$	$69.9\,[51.2, 116]$

Table 6: First matched Stage-2 $B_{s1} \to B_s\gamma$ baseline. Brackets show the range over the Borel and continuum-threshold grid, not a full uncertainty interval. The B_{s1} decay constants are working heavy-quark-scaled inputs, so these values should be refined after dedicated B_{s1} hadronic inputs are fixed.

The qualitative behavior mirrors the D_{s1} calculation: the observed high-combination state is cancellation-sensitive, while the lower-combination diagnostic is much larger. The difference between the legacy and lattice photon-normalization rows is again substantial, so the bottom-sector analysis should keep both scenarios until the input convention is finalized. The machine-readable grid is `outputs/bs1_stage2_baseline.csv`.

The companion script

`scripts/bs1_uncertainty_and_plots.py`

performs a 500-point Monte Carlo scan for each state, photon-input scenario, and mixing-angle treatment. The observed $B_{s1}(5830)$ row samples $8 \le M^2 \le 14$ GeV^2, $6.05 \le \sqrt{s_0} \le 6.25$ GeV, the same quark and photon-DA inputs used in the D_{s1} scan, and assigns a conservative 20% width to the temporary heavy-quark-scaled B_{s1} axial and tensor decay constants. The lower-state diagnostic uses the lattice-inspired model mass $m_{B_{s1}} = 5.750 \pm 0.026$ GeV from Lang, Mohler, Prelovsek and Woloshyn.

After the physical-current normalization discussion above, the temporary HQ-scaled B_{s1} constants are not the preferred bottom-sector input. The script

`scripts/stage3_bs1_pz_decay_constants.py`

therefore reruns the same B_{s1} Monte Carlo scan with the Pullin–Zwicky heavy-light sum-rule inputs $f_{B_{s1}} = 0.305 \pm 0.027$ GeV and $f^T_{B_{s1}} = 0.285 \pm 0.027$ GeV. This is not yet a full mixed-current Π_{AA}–Π_{BB}–Π_{AB} two-point solution, but it replaces the naive HQ-scaled decay constants by dedicated bottom-sector hadronic inputs. The table below is therefore the preferred B_{s1} scenario at this stage.

The most important lesson from Table 7 is that the observed $B_{s1}(5830)$ amplitude can pass close to zero over the accepted input range. With the PZ decay constants the observed-state width is naturally sub-keV in both photon-input scenarios, whereas the lower-partner diagnostic remains at the tens-of-keV level. The machine-readable Stage-3 bottom outputs are `outputs/stage3_bs1_pz_decay_constant_scan.csv` and `outputs/stage3_bs1_pz_decay_constant_summar` the earlier `outputs/bs1_stage2_mc_summary.csv` file is retained as the HQ-scaled diagnostic.

Target	Photon input	Mixing input	Decay constants	G (GeV^{-1})	Γ (keV)
$B_{s1}(5830)$ high combo	legacy χ_s (Rohrwild/BBK)	$\theta = 35.3°$	PZ	$-0.0140\,[-0.0387, +0.0155]$	$0.109\,[0.008, 0.413]$
$B_{s1}(5830)$ high combo	legacy χ_s (Rohrwild/BBK)	$25° \leq \theta \leq 45°$	PZ	$-0.0126\,[-0.0527, +0.0364]$	$0.249\,[0.023, 0.819]$
$B_{s1}(5830)$ high combo	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$\theta = 35.3°$	PZ	$+0.0356\,[+0.0029, +0.0694]$	$0.272\,[0.028, 1.024]$
$B_{s1}(5830)$ high combo	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$25° \leq \theta \leq 45°$	PZ	$+0.0330\,[-0.0228, +0.0974]$	$0.429\,[0.028, 2.094]$
lower 1^+ diagnostic	legacy χ_s (Rohrwild/BBK)	$\theta = 35.3°$	PZ	$+0.337\,[+0.279, +0.411]$	$14.2\,[8.93, 21.2]$
lower 1^+ diagnostic	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$\theta = 35.3°$	PZ	$+0.474\,[+0.420, +0.546]$	$27.8\,[20.1, 39.4]$

Table 7: Monte Carlo $B_{s1} \to B_s\gamma$ scan using the Stage-3 Pullin–Zwicky decay-constant scenario. Entries are medians with 16–84 percentile intervals. The observed $B_{s1}(5830)$ amplitude is cancellation-sensitive, so the width should be quoted with asymmetric percentile intervals rather than a Gaussian error.

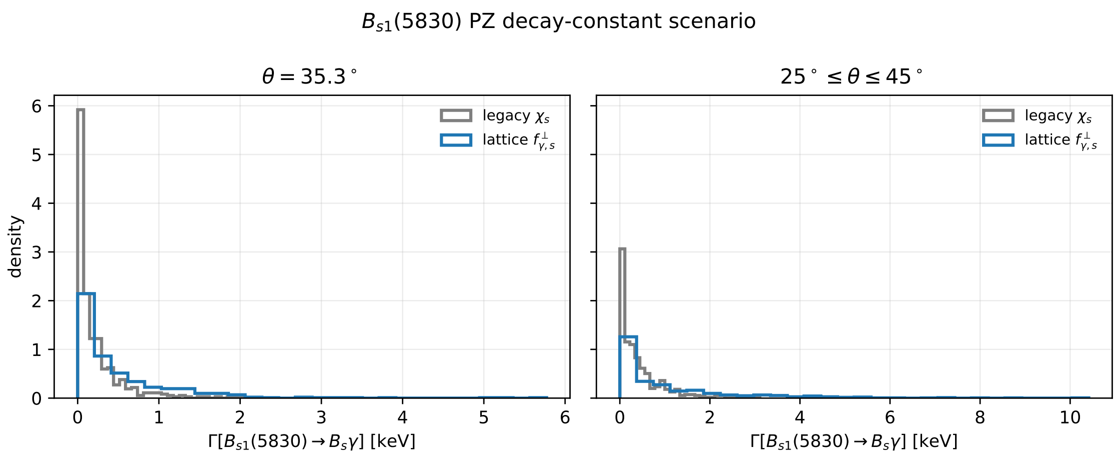

Figure 12: Monte Carlo width distributions for the observed $B_{s1}(5830) \to B_s\gamma$ high-combination state in the PZ decay-constant scenario. The legacy and lattice-normalized photon inputs are shown separately for fixed $\theta = 35.3°$ and for the mixing-angle scan.

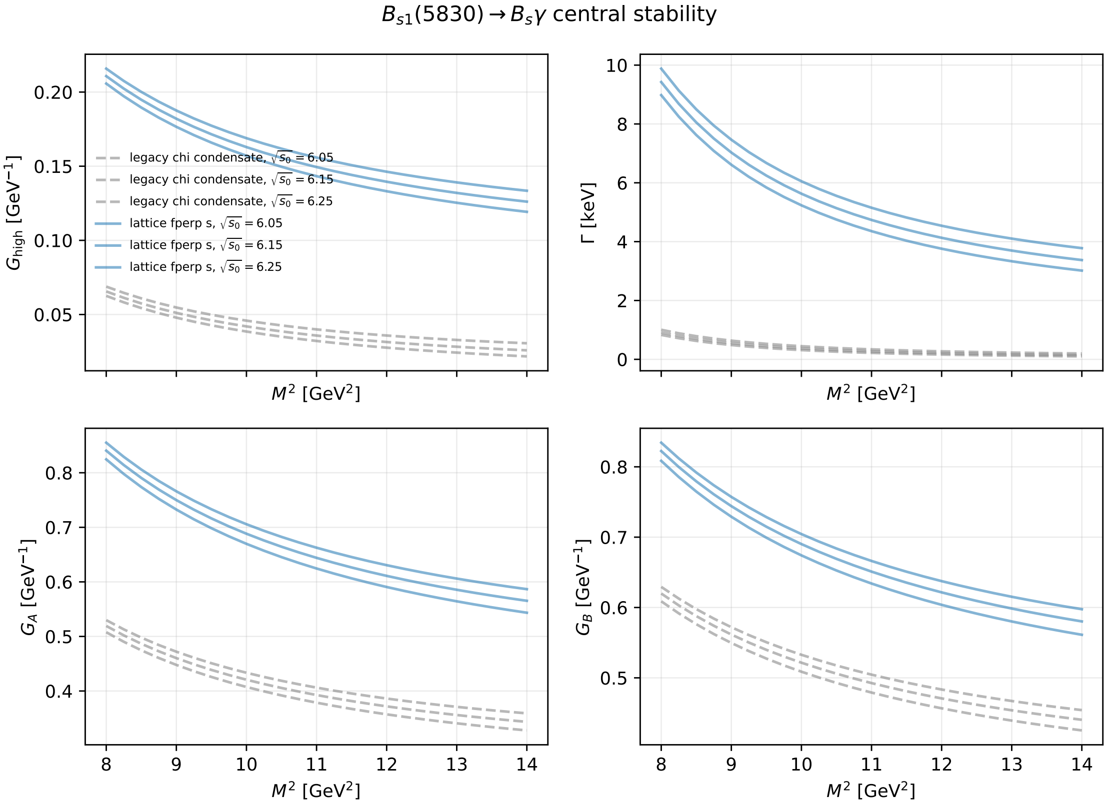

Figure 13: Central $B_{s1}(5830) \to B_s\gamma$ stability curves for the high-combination amplitude, width, and unmixed G_A, G_B components.

The same warning about current normalization applies in the bottom sector, but there we do not presently have an analogue of the charm Stage-3 two-point calibration. The PZ inputs test the axial and tensor basis-current scales, not a fully derived mixed-current residue matrix. Consequently the bottom quantity denoted $f_{\rm eff}$ in the combined tables should be read as the sampled effective normalization entering the quoted high- or low-combination amplitude. It is not an independently extracted local two-point residue f_1 or f_2. A future bottom two-point calculation would require the complete AA, BB, and AB correlator matrix, exactly as in the charm discussion above.

21 Combined D_{s1} and B_{s1} Summary

Table 8 collects the current width predictions in one place. The D_{s1} rows include the Stage-3 physical-current residue normalization described above. The B_{s1} rows use the Stage-3 Pullin–Zwicky decay-constant scenario. In addition to the observed $B_{s1}(5830)$ high-combination state, the table now includes the lower 1^+ bottom-strange diagnostic state used for comparison with the $B_{s1}(5748)$ channel in Bondar–Milstein's phenomenology. This lower row is not an experimentally established resonance. All entries are medians with 16–84 percentile intervals. The table is intentionally explicit about the photon input, mixing input, normalization status, normalization constants, form factor G, and width Γ, so that no reader has to infer which scenario is being quoted.

Sector	Decay	Photon input	Mixing input	Normalization status	Normalization input	G (GeV^{-1})	Γ (keV)
D_s final	$D_{s1}(2460) \to D_s\gamma$	legacy χ_s (Rohrwild/BBK)	$\theta = 35.3°$	Stage-3 residue calibrated	$f_1 = 0.344[0.329, 0.363]$, $f_2 = 0.380[0.339, 0.423]$	$-0.424\,[-0.484, -0.360]$	$37.7^{+11.5}_{-10.5}$
D_s final	$D_{s1}(2536) \to D_s\gamma$	legacy χ_s (Rohrwild/BBK)	$\theta = 35.3°$	Stage-3 residue calibrated	$f_1 = 0.344[0.329, 0.363]$, $f_2 = 0.380[0.339, 0.423]$	$+0.0118\,[+0.00143, +0.0233]$	$0.045^{+0.124}_{-0.040}$
D_s final	$D_{s1}(2460) \to D_s\gamma$	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$\theta = 35.3°$	Stage-3 residue calibrated	$f_1 = 0.345[0.330, 0.364]$, $f_2 = 0.379[0.338, 0.423]$	$-0.277\,[-0.312, -0.243]$	$16.1^{+4.4}_{-3.7}$
D_s final	$D_{s1}(2536) \to D_s\gamma$	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$\theta = 35.3°$	Stage-3 residue calibrated	$f_1 = 0.345[0.330, 0.364]$, $f_2 = 0.379[0.338, 0.423]$	$+0.0403\,[+0.0194, +0.0596]$	$0.504^{+0.600}_{-0.377}$
B_s final	$B_{s1}(5830) \to B_s\gamma$	legacy χ_s (Rohrwild/BBK)	$\theta = 35.3°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.391[0.354, 0.428]$	$-0.0140\,[-0.0387, +0.0155]$	$0.109^{+0.304}_{-0.101}$
B_s final	$B_{s1}(5830) \to B_s\gamma$	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$\theta = 35.3°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.393[0.354, 0.427]$	$+0.0356\,[+0.0029, +0.0694]$	$0.272^{+0.752}_{-0.244}$
B_s final	$B_{s1}(5830) \to B_s\gamma$	legacy χ_s (Rohrwild/BBK)	$25° \le \theta \le 45°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.394[0.356, 0.430]$	$-0.0126\,[-0.0527, +0.0364]$	$0.249^{+0.570}_{-0.226}$
B_s final	$B_{s1}(5830) \to B_s\gamma$	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$25° \le \theta \le 45°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.387[0.348, 0.423]$	$+0.0330\,[-0.0228, +0.0974]$	$0.429^{+1.665}_{-0.401}$
B_s final	$B_{s1}^{\rm low}(5750) \to B_s\gamma$	legacy χ_s (Rohrwild/BBK)	$\theta = 35.3°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.384[0.347, 0.421]$	$+0.337\,[+0.279, +0.411]$	$14.2^{+7.0}_{-5.2}$
B_s final	$B_{s1}^{\rm low}(5750) \to B_s\gamma$	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$\theta = 35.3°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.386[0.350, 0.424]$	$+0.474\,[+0.420, +0.546]$	$27.8^{+11.6}_{-7.7}$
B_s final	$B_{s1}^{\rm low}(5750) \to B_s\gamma$	legacy χ_s (Rohrwild/BBK)	$25° \le \theta \le 45°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.387[0.348, 0.426]$	$+0.325\,[+0.274, +0.400]$	$13.0^{+8.7}_{-4.2}$
B_s final	$B_{s1}^{\rm low}(5750) \to B_s\gamma$	lattice $f^{\perp}_{\gamma,s}$ (Bacchio et al.)	$25° \le \theta \le 45°$	PZ decay constants	$f_A^{\rm PZ} = 0.305(27)$, $f_T^{\rm PZ} = 0.285(27)$; $f_{\rm eff} = 0.380[0.349, 0.419]$	$+0.468\,[+0.416, +0.541]$	$26.7^{+10.9}_{-7.3}$

Table 8: Current combined summary of D_{s1} and B_{s1} radiative form factors and widths. The G column is the E1 coupling in the convention of Eq. (37). The D_s rows include the Stage-3 physical-current residue correction. The B_s rows use the Stage-3 Pullin–Zwicky decay-constant scenario; $f_{\rm eff}$ is the sampled effective mixed-current residue entering the quoted B_{s1} amplitude, not an independently derived local two-point f_1 or f_2. The $B_{s1}^{\rm low}(5750)$ rows are lower-partner diagnostics, not observed-state measurements. The older HQ-scaled bottom rows are retained only as diagnostics.

The machine-readable version of Table 8 is `outputs/combined_recommended_results_table.csv`.

21.1 Comparison With Bondar–Milstein Literature Tables

Bondar and Milstein collect several quark-model predictions for the D_{s1} and B_{s1} radiative widths. Their charm table uses Godfrey, Goity–Roberts, Green–Repko–Radford, Radford–Repko–Saelim, Chen–Liu–Zhou–Chen, and Korner–Pirjol–Schilcher for the comparison columns. Their bottom table uses Li–Ni–Zhong, Godfrey–Moats–Swanson, and Lu–Pan–Wang–Wang–Li. We cite these papers explicitly in the manuscript, because otherwise the comparison would look like a citation only to Bondar–Milstein's summary table.

Table 9 gives a compact version restricted to channels computed in the present LCSR analysis. The comparison set spans early quark-model estimates, relativistic quark-model calculations, more recent potential-model/electric-transition analyses, and the Bondar–Milstein phenomenological extraction. The full machine-readable table also stores the $D_s^*\gamma$ and $B_s^*\gamma$ rows from the literature; our entry is marked as not computed for those channels because they require separate vector-final-state correlators. This scope statement should be kept in the paper so that the comparison table is not read as a claim about uncalculated vector-final-state transitions.

Channel	Early QM	Rel. QM	Recent QM	Bondar–Milstein	This work
$D_{s1}(2536) \to D_s\gamma$	15 (Godfrey2005)	25.2–31.1 (Goity–Roberts2001)	18.18–18.85 (Chen2020)	12 (Bondar–Milstein2025)	$0.504[0.127, 1.104]$
$D_{s1}(2460) \to D_s\gamma$	6.2 (Godfrey2005)	10.3–17.2 (Goity–Roberts2001)	3.53–3.61 (Chen2020)	297 (Bondar–Milstein2025)	$16.1[12.4, 20.5]$
$B_{s1}^{\rm low}(5750) \to B_s\gamma$	37 (Li–Ni–Zhong2021)	70.6 (Godfrey–Moats–Swanson2016)	97.7 (Lu–Pan–Wang2016)	47 (Bondar–Milstein2025)	$27.8[20.1, 39.4]$
$B_{s1}(5830) \to B_s\gamma$	27 (Li–Ni–Zhong2021)	47.8 (Godfrey–Moats–Swanson2016)	56.6 (Lu–Pan–Wang2016)	28 (Bondar–Milstein2025)	$0.272[0.028, 1.024]$

Table 9: Representative comparison with the radiative-width literature summarized by Bondar–Milstein. Widths are in keV. The entries in the last column use our preferred lattice photon normalization. The complete bookkeeping table is `outputs/bondar_literature_comparison_table.csv`.

22 Comparison With Experimental Information

The sum rule predicts a partial radiative width, $\Gamma_\gamma \equiv \Gamma(H_1 \to H\gamma)$, while experiments often quote branching fractions, total widths, or spectroscopy measurements in other decay channels. Therefore the comparison is not always one-to-one. In Table 10, the "interpretation" column has the following meaning:

- for $D_{s1}(2460)$, the measured branching fraction is used to infer the total-width scale that would be implied by our partial radiative width;
- for $D_{s1}(2536)$, the measured total width is used to infer the radiative branching fraction implied by our partial width;
- for $B_{s1}(5830)$, no direct radiative measurement is available, so the row is a prediction for future experimental searches rather than a numerical comparison.

Decay	Theory input	Experimental information	Interpretation
$D_{s1}(2460) \to D_s\gamma$	legacy χ_s: $37.7\,[27.2, 49.2]$ keV	$\mathcal{B}(D_{s1}(2460) \to D_s\gamma) = (16 \pm 4 \pm 3)\%$ (BaBar)	implies $\Gamma_{\rm tot} \sim 0.24$ MeV; compatible with a very narrow state
$D_{s1}(2460) \to D_s\gamma$	lattice $f_{\gamma,s}^{\perp}$: $16.1\,[12.4, 20.5]$ keV	$\mathcal{B}(D_{s1}(2460) \to D_s\gamma) = (16 \pm 4 \pm 3)\%$ (BaBar)	implies $\Gamma_{\rm tot} \sim 0.10$ MeV; also compatible with a very narrow state
$D_{s1}(2536) \to D_s\gamma$	legacy χ_s: $0.045\,[0.0046, 0.169]$ keV	$\Gamma_{\rm tot}(D_{s1}(2536)) = 0.92 \pm 0.05$ MeV (BaBar)	implies $\mathcal{B}_\gamma \sim 0.005\%$; extremely cancellation suppressed
$D_{s1}(2536) \to D_s\gamma$	lattice $f_{\gamma,s}^{\perp}$: $0.504\,[0.127, 1.104]$ keV	$\Gamma_{\rm tot}(D_{s1}(2536)) = 0.92 \pm 0.05$ MeV (BaBar)	implies $\mathcal{B}_\gamma \sim 0.05\%$; strongly cancellation suppressed
$B_{s1}(5830) \to B_s\gamma$	legacy χ_s: $0.109\,[0.008, 0.413]$ keV	no direct radiative measurement; observed mainly in $B^{(*)}K$ spectroscopy	prediction for future searches; naturally sub-keV in the PZ scenario
$B_{s1}(5830) \to B_s\gamma$	lattice $f_{\gamma,s}^{\perp}$: $0.272\,[0.028, 1.024]$ keV	no direct radiative measurement; observed mainly in $B^{(*)}K$ spectroscopy	prediction for future searches; high-combination amplitude remains cancellation sensitive

Table 10: Comparison of the present radiative-width predictions with available experimental information. The BaBar $D_{s1}(2460)$ branching fraction is from the $B \to D^{(*)}D_{sJ}^{(*)}$ analysis. The $D_{s1}(2536)$ total width is the BaBar width measurement. The $B_{s1}(5830)$ has no direct radiative branching-fraction measurement at present; CMS and LHCb measurements are spectroscopy measurements in hadronic channels.

Numerically, the $D_{s1}(2460)$ rows say that the present partial-width prediction is consistent with a total width well below the MeV scale if the measured radiative branching fraction is used. The $D_{s1}(2536)$ rows say something sharper: with the measured total width near 0.92 MeV, the radiative branching fraction is expected to be far below the percent level. For $B_{s1}(5830)$ the table should be read as a target for future experimental searches, not as a test against an existing radiative measurement.

The machine-readable version of Table 10 is `outputs/experimental_comparison_table.csv`.

23 Paper-Ready Ingredients Before Writing

This section collects the compact version of the calculation choices that should be carried into the manuscript. It is deliberately shorter than the derivation above: the purpose is to state what a reader must know before looking at the final numerical tables.

23.1 Compact Input Table

Quantity	Symbol	Value	Comment
charm mass	m_c	1.27 ± 0.02 GeV	$\overline{\mathrm{MS}}$ input for D_s sector; PDG2024
bottom mass	m_b	4.18 ± 0.03 GeV	$\overline{\mathrm{MS}}$ input for B_s sector; PDG2024
strange mass	m_s	0.093 ± 0.011 GeV	retained explicitly, including m_s^2 terms; PDG2024
$D_{s1}(2460), D_{s1}(2536), D_s$ masses	$M_{2460}, M_{2536}, m_{D_s}$	$2.4595,\ 2.53511,\ 1.96835$ GeV	physical charm-strange masses; PDG2024
$B_{s1}(5830), B_s$ masses	M_{5830}, m_{B_s}	$5.82870,\ 5.36692$ GeV	observed bottom-strange channel; PDG2024/CMS2018
lower B_{s1} diagnostic mass	$M_{B_{s1}}^{\mathrm{low}}$	5.750 ± 0.026 GeV	expected, not-yet-established lower 1^+ partner; Lang2015
D_s and D_{s1} decay constants	$f_{D_s}, f_{D_{s1},A}, f_{D_{s1}}^T$	$0.2499,\ 0.225,\ 0.256$ GeV	hadronic-side inputs; FLAG2024/Colangelo2005/PullinZwicky2021
D_{s1} physical-current residues	$f_1,\ f_2$	$0.345 \pm 0.017,\ \simeq 0.38$ GeV	working Stage-3 anchor; f_2 from overlap calibration; checked against Wang2015 pure axial benchmark
B_s and working B_{s1} decay constants	$f_{B_s}, f_{B_{s1},A}, f_{B_{s1}}^T$	$0.2303,\ 0.146,\ 0.164$ GeV	diagnostic HQ-scaled inputs, retained only as a cross-check
PZ B_{s1} decay-constant scenario	$f_{B_{s1},A}^{\mathrm{PZ}}, f_{B_{s1},T}^{\mathrm{PZ}}$	$0.305 \pm 0.027,\ 0.285 \pm 0.027$ GeV	dedicated bottom-sector input scenario; PullinZwicky2021
condensates	$\langle \bar{q}q \rangle, \kappa$	$-(0.240\ \mathrm{GeV})^3,\ 0.8 \pm 0.1$	$\langle \bar{s}s \rangle = \kappa \langle \bar{q}q \rangle$; FLAG2024/standard SVZ input
legacy photon input	$\chi_s(1\ \mathrm{GeV})$	$+3.15 \pm 0.30\ \mathrm{GeV}^{-2}$	Rohrwild/BBK-style comparison scenario; BBK2002/Rohrwild2007
lattice photon input	$f_{\gamma,s}^{\perp}(1\ \mathrm{GeV})$	-55.1 ± 1.9 MeV	preferred leading-twist normalization scenario; Bacchio2024
mixing angle	θ	35.3° central, 25°–45° scan	${}^3P_1$–${}^1P_1$ physical-state mixing; Colangelo2005 convention
D_s sum-rule window	$M^2, \sqrt{s_0}$	3–6 GeV^2, 2.50–2.60 GeV	charm-sector stability/uncertainty scan
B_s sum-rule window	$M^2, \sqrt{s_0}$	8–14 GeV^2, 6.05–6.25 GeV	observed $B_{s1}(5830)$ scan

Table 11: Compact input table for the manuscript. The complete machine-readable version is `outputs/paper_input_summary_table.csv`; the longer bookkeeping table is `inputs_table.csv`.

23.2 Convention Checklist For The Paper

- We use the mostly-minus metric and $\epsilon^{0123} = +1$.
- The initial axial-meson momentum is $P = p + q$; p is the outgoing pseudoscalar momentum and q is the photon momentum.
- The tensor current is $\mathrm{i}\, \bar{s}\sigma_{\mu\alpha} P^\alpha \gamma_5 Q/(m_Q + m_s)$, so the momentum entering the tensor current is the initial-state momentum P.
- The physical amplitudes are $G_{\mathrm{low}} = \sin\theta\, G_A + \cos\theta\, G_B$ and $G_{\mathrm{high}} = \cos\theta\, G_A - \sin\theta\, G_B$ in the basis-current normalization convention.
- For the physical currents J_1 and J_2, the final couplings should be divided by the corresponding two-point residues f_1 and f_2 as in Eqs. (62)–(63).
- The width is computed from $\Gamma = \alpha_{\mathrm{em}} G^2 |\vec{q}_\gamma|^3/3$.
- We store $\chi_s(1\ \mathrm{GeV})$ as a positive Rohrwild/BBK magnitude, $\chi_s = +3.15\ \mathrm{GeV}^{-2}$; this must not be mixed with papers that print the opposite sign convention.

23.3 Propagator And Condensate Statement

The base propagators in the hard calculation are not expanded in ordinary vacuum condensates. Instead, we use the quark propagators in an external electromagnetic field for hard photon emission and the background-gluon propagator term for the three-particle photon DAs. The condensates that appear in the final formulas are therefore part of the photon matrix elements and photon DA normalizations, such as $\chi_s \langle \bar{s}s \rangle$, not separate SVZ condensate corrections to the free propagator. This distinction should be stated clearly in the paper because it is one of the main convention choices of the calculation.

23.4 Quoting Strategy

For the final paper tables, the lattice-normalized $f_{\gamma,s}^{\perp}$ scenario should be presented as the preferred modern input choice, while the legacy $\chi_s \langle \bar{s}s \rangle$ scenario should be kept for comparison with the older photon-DA literature. Uncertainties should be quoted as medians with 16–84 percentile intervals. This is essential for $D_{s1}(2536)$ and $B_{s1}(5830)$ because their amplitudes can pass close to zero, making Gaussian errors misleading.

For the bottom-strange sector, the observed $B_{s1}(5830)$ prediction is a future-search target. The lower B_{s1} row is a prediction for an expected but not yet experimentally established 1^+ state; it should not be described as an observed resonance. Its role in the manuscript is to show that the lower combination is constructive and can have a much larger radiative width than the observed high-combination state.

23.5 Reference Map

The manuscript reference list should connect each input family to its source: PDG for masses and quark charges, FLAG Review 2024 for f_{B_s} and related lattice decay-constant context, Rohrwild/BBK for the legacy photon DAs, Bacchio et al. for the lattice $f^{\perp}_{\gamma,s}$ normalization, Colangelo, De Fazio and Ozpineci for the original charm-strange LCSR benchmark, Pullin and Zwicky for heavy-light decay-constant cross-checks, Lang–Mohler–Prelovsek– Woloshyn for the lower B_{s1} diagnostic mass, BaBar for the $D_{s1}(2460)$ branching fraction and $D_{s1}(2536)$ total width, and CMS/LHCb for the observed $B_{s1}(5830)$ spectroscopy. The Bondar and Bondar–Milstein papers should be cited in the discussion of the $D_{s1}(2536)$ nonobservation, the constructive/destructive E1-amplitude mechanism, and the comparison with the lower and higher B_{s1} radiative channels.

Citation key	Input family	Used for
PDG2024	masses, quark masses, charges, α_{em}	kinematics, quark-mass inputs, width normalization
FLAG2024	lattice decay constants and condensate context	f_{D_s}, f_{B_s}, and modern lattice normalization context
Colangelo2005	charm-strange radiative LCSR benchmark	comparison calculation and charm axial-current normalization
BBK2002	photon distribution amplitudes	twist-2, twist-3, and twist-4 photon DA shapes
Rohrwild2007	magnetic susceptibility update	legacy χ_s comparison scenario
Bacchio2024	lattice photon-DA normalization	preferred $f^{\perp}_{\gamma,s}(1\ \text{GeV})$ scenario
PullinZwicky2021	heavy-light vector/tensor decay constants	D_s tensor scale cross-check and preferred B_{s1} PZ scenario
Wang2015	heavy-light decay constants from QCDSR	pure axial-current $f_A(D_{s1})$ benchmark for the Stage-3 residue check
Lang2015	positive-parity B_s spectroscopy	lower B_{s1} diagnostic mass
BaBar2006	$D_{s1}(2460)$ branching data	experimental interpretation of $\Gamma(D_{s1}(2460) \to D_s\gamma)$
BaBar2011	$D_{s1}(2536)$ total width	branching-fraction estimate for the high charm-strange state
CMS2018	$B_{s1}(5830)$ spectroscopy	observed bottom-strange mass and phenomenological comparison
Bondar2025Why	$D_{s1}(2536)$ radiative nonobservation	experimental upper-limit motivation for the suppressed high-state mode
BondarMilstein2025	D_{s1} and B_{s1} radiative phenomenology	constructive/destructive E1-amplitude mechanism and comparison tables
Godfrey2005, GoityRoberts2001, Chen2020	D_{s1} radiative-width literature	comparison columns for $D_{s1} \to D_s^{(*)}\gamma$
LiNiZhong2021, GodfreyMoatsSwanson2016, LuPanWang2016	B_{s1} radiative-width literature	comparison columns for $B_{s1} \to B_s^{(*)}\gamma$

Table 12: Citation map for the numerical inputs and comparison data. The machine-readable version is `outputs/input_citation_map.csv`.

24 Pre-Paper Calculation Closure Checklist

The core D_{s1} and B_{s1} calculation set is complete at the precision level used in these working notes. The completed items are:

- Stage 1 hard-photon and two-particle photon-DA terms, with both heavy- and strange-line hard emission included;
- Stage 2 three-particle photon-DA corrections from the background-gluon insertion in the quark propagator;
- explicit retention of the m_s and m_s^2 terms used by the present Mathematica/Python kernels;
- D_{s1} Stage 3 physical-current normalization using the calibrated residues $f_1 \simeq 0.345$ GeV and $f_2 \simeq 0.38$ GeV, while using Wang2015 only as a pure-axial benchmark rather than as a direct authority for the mixed residues;
- B_{s1} Stage 3 normalization with the Pullin–Zwicky decay-constant scenario, plus the lower-state diagnostic channel;

- one-at-a-time uncertainty diagnostics, Monte Carlo uncertainty scans, percentile intervals, stability plots in (M^2, s_0), and comparison tables;
- experimental interpretation tables for the available D_{s1} and B_{s1} information;
- citation-tagged input tables in `inputs_table.csv`, `outputs/paper_input_summary_table.csv`, and `outputs/input_citation_map.csv`.

The only non-blocking physics refinement left before submission is an independent local two-point validation of the complete current matrix:

- derive the local two-point OPE for Π_{AA}, Π_{BB} and Π_{AB}, including the ordinary SVZ condensates appropriate to a two-point decay-constant sum rule;
- compare the extracted f_1 and f_2 with the Stage-3 calibrated values, $f_1 \simeq 0.345$ GeV and $f_2 \simeq 0.38$ GeV;
- rerun the D_{s1} headline table only if the independent two-point residues differ appreciably from the calibrated values;
- optionally repeat the same fully mixed two-point check in the bottom-strange sector once a dedicated local-QCDSR normalization is desired.

Private future-work note. The B_{c1} radiative system should not be included in the present paper. It is a heavy-heavy problem rather than a heavy-light strange-meson problem, so the present photon-DA LCSR machinery does not carry over without a new formalism. We keep this only as an internal next-study idea in `notes/internal_future_work.md`.

25 Remaining Manuscript Tasks

The remaining tasks are now writing and presentation tasks rather than missing core calculations:

- decide journal style and create the manuscript source file;
- convert the working-note equations into a shorter derivation section;
- select only the final numerical tables and the most useful stability, uncertainty, and comparison figures;
- build a BibTeX file and cite every input in Tables 11 and 12;
- decide whether the B_{s1} predictions remain in the main text or in a separate phenomenology subsection; the preferred rows now use the Pullin–Zwicky decay-constant scenario, while a fully mixed bottom-sector two-point matrix is best treated as a validation refinement.